STANDARD OPERATING PROCEDURES

OFP C172 FR

Flight Planning Application
User Manual

DOCUMENT	OFP-SOP – Application User Manual
REVISION	Rev. 3 – DRAFT for review (new interface, Documents page, Navlog profile)
DATE	07 NOV 2026
AIRCRAFT	Reims Cessna “Reims Rocket” – I-CCAF (FR172J), I-CCAB (FR172H)
OPERATIONS	VFR flight planning
OUTPUTS	Operational Flight Plan (OFP, A4) · NAV-FLIGHTPLAN (A5) · Aeroclub Varese navlog (2 × A4)

> This manual describes how to operate the application. It does not replace the Aircraft Flight Manual, the Operations Manual or the flight school SOPs. The documents produced are an aid to planning: responsibility for the flight remains with the Pilot in Command.

Contents

Conventions 3

Abbreviations 3

SECTION 1 General 5

1.1 Purpose 5

1.2 Scope and applicability 5

1.3 System overview 5

1.4 Data sources 5

1.5 Responsibilities 5

1.6 Data storage 6

SECTION 2 Data Ingestion – Import Page 7

2.1 Pre-requisites 7

2.2 Procedure 7

SECTION 3 Pre-flight Data Entry – Wizard 10

3.0 Wizard navigation 10

3.1 Step 1 – Flight & Aircraft 11

3.2 Step 2 – Route & Fuel 13

3.3 Step 3 – Mass & Balance 17

3.4 Step 4 – Performance 18

3.5 Step 5 – Navlog 20

3.6 Step 6 – Destination Charts 22

3.7 Step 7 – Briefing 22

3.8 Step 8 – NOTAM & Weather 24

SECTION 4 System Outputs & Dispatch 26

4.1 Documents 26

4.2 Menu 27

4.3 Operational Flight Plan (A4) 27

4.4 NAV-FLIGHTPLAN (A5) and club navlog 31

4.5 Dashboard – Saved flights 34

SECTION 5 Operational Notes – Safety & Performance Limits 36

5.1 Aircraft applicability 36

5.2 Performance limits 36

5.3 Fuel planning as implemented 36

5.4 Navigation data 37

5.5 Daylight check 37

5.6 Documents and storage 37

APPENDIX A Pre-Departure Cross-Check 38

Conventions

PROCEDURE VERBS

LOAD	Upload a file (drag and drop or tap the drop zone).
INPUT	Type a value.
SELECT	Choose from a list, tab, tick box or menu.
VERIFY	Check that a value filled in by the application is present and plausible.
CROSS-CHECK	Compare a value with an independent source (AIP, AFM, briefing, ForeFlight).
CORRECT	Overwrite a value filled in by the application.
REVIEW	Read every message shown and act on each one.

FIELD CLASSIFICATION

M MANDATORY	Pilot input required. Without it the related output is missing or wrong.
A AUTO	Imported or computed by the application. Editable unless shown greyed out. Must be VERIFIED.
O OPTIONAL	Not required for a valid output.

The application does not block progress on empty fields. The classification above is the operating standard of this manual, not a software lock.

FIGURES AND CALLOUTS

Figures are taken from the application in the light theme, with a real flight (LILN – T&G LILE – LILN, alternate LIMF). Lettered markers **A** **B** … identify the elements described in the legend under each figure and in the procedure that cites it. The figure button ⤢ enlarges a figure on screen.

NOTES, CAUTIONS AND WARNINGS

WARNING
Could affect the safety of the flight if ignored.

CAUTION
Could produce a wrong or incomplete document.

NOTE
Additional information.

ABBREVIATIONS

AFM	Aircraft Flight Manual	**AIP**	Aeronautical Information Publication
ALTN	Alternate aerodrome	**ARR / DEST**	Arrival / destination aerodrome
ATO	Actual time over	**BEW**	Basic empty weight
CG	Centre of gravity	**C/S**	Callsign
DA / PA	Density / pressure altitude	**DEP**	Departure aerodrome
ECT	End of evening civil twilight	**EET**	Estimated elapsed time (leg)
ETO	Estimated time over	**FF**	Fuel flow
FL	Flight level	**FOB**	Fuel on board
GS	Ground speed	**LDA**	Landing distance available
MAP	Manifold pressure	**MLAW / MTOW**	Maximum landing / take-off weight

OFP	Operational flight plan	**PIC**	Pilot in command
POB	Persons on board	**POH**	Pilot's Operating Handbook
SR / SS	Sunrise / sunset	**T&G**	Touch and go
TAS	True airspeed	**TEM**	Threat and error management
TOC	Top of climb	**TOD**	Top of descent
TODA / TORA	Take-off distance / run available	**TOW**	Take-off weight
UTC	Coordinated universal time	**ZFW**	Zero fuel weight

SECTION 1 General

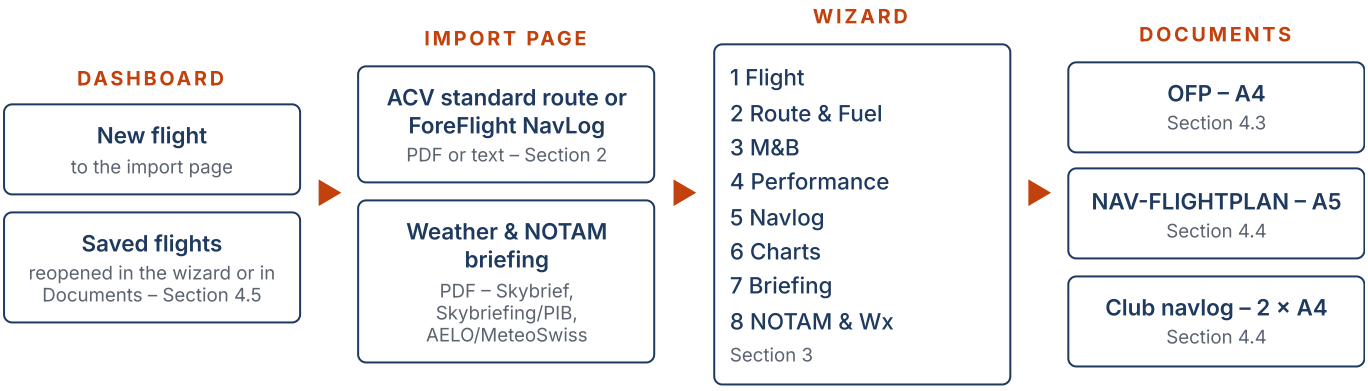
1.1 Purpose

The application converts the route prepared in ForeFlight and the weather briefing into the two documents carried in flight: the **Operational Flight Plan (OFP)** and the **NAV-FLIGHTPLAN**. All performance, fuel and mass figures are computed from the AFM of the selected aircraft; every intermediate result is shown so that it can be checked.

1.2 Scope and applicability

- Aircraft: Reims Cessna "Reims Rocket", Continental IO-360-D: **I-CCAF** (FR172J, AFM Edition 3) and **I-CCAB** (FR172H, AFM Edition 2). The two AFMs differ only in VR, VY, stall speeds and VA, and the club gives a different VX for each (I-CCAF 68 MPH, I-CCAB 75 MPH): the application uses those of the aircraft selected in Step 1.
- Operations: VFR flight planning. Final reserve is selectable for VFR (30 min) or IFR (45 min).
- Units: fuel in **litres**, fuel flow in **L/h**, speeds in **MPH** (as in the AFM), distances in **NM**, times in **UTC**.

1.3 System overview



1.4 Data sources

DATA	SOURCE
Cruise power, TAS, fuel flow	AFM fig. 5-1, interpolated between 2500 / 5000 / 7500 / 10000 / 15000 ft
Climb time, fuel (incl. start-up and take-off)	AFM fig. 5-6, 1157 kg; climb distance derived from the same data
Take-off / landing distances	AFM fig. 5-4 / 5-5
Basic empty weight, moments, arms	Flying club W. & B. Loading Form (Rev. 17)
Aerodrome position, elevation, runways, frequencies	OurAirports database (physical runway length only)
Route, checkpoints, tracks, distances	ForeFlight NavLog
METAR, SPECI, TAF, NOTAM, SIGMET/AIRMET, charts	Weather briefing PDF

1.5 Responsibilities

The PIC is responsible for the correctness of every input, for every cross-check required by this manual, and for the decision to fly. The application does not navigate, does not decide and does not replace the AFM or the AIP.

1.6 Data storage

Access is restricted: accounts are created by the administrator (Section 2.1). Flights are stored only when the pilot uses **Save** or **Proceed** (Sections 3.0 and 3.8). They are saved to the pilot's account, are available on any device after signing in and are not visible to other users. A copy is kept on the device in use, so work can continue without a connection: changes are sent when the connection is restored.

SECTION 2 Data Ingestion – Import Page

2.1 Pre-requisites

1. **INPUT** the e-mail and password provided by the administrator to sign in. At the first sign-in, replace the temporary password with your own (at least 10 characters). The **Dashboard** opens (Section 4.5).
2. **PREPARE** the route in ForeFlight and export the **NavLog** (PDF or text), unless an **ACV standard route** is flown.
3. **DOWNLOAD** the **weather and NOTAM briefing** PDF covering departure, destination, alternate and route, from SkyBrief (link in the Weather Briefing card).
4. **SELECT** **New flight** on the Dashboard, or **+** in the side bar: the Import page opens. **Flights** in the side bar goes back without importing.

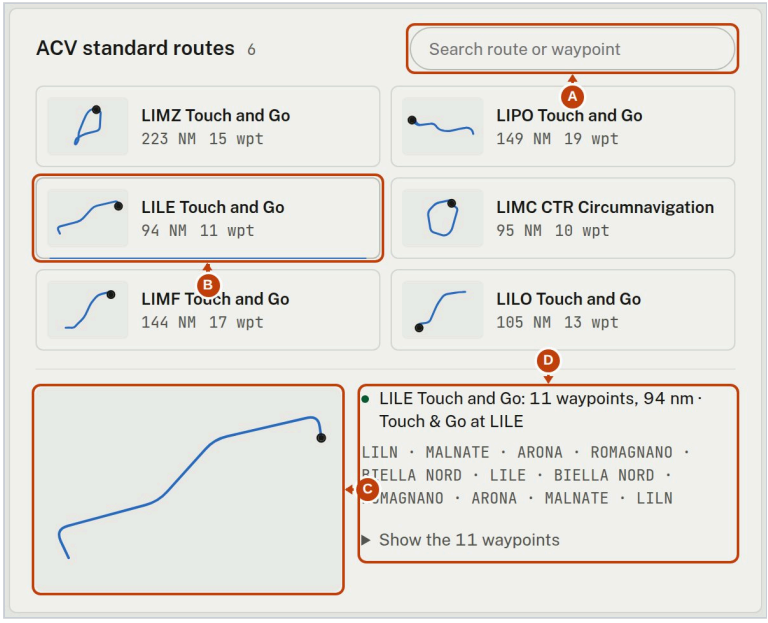
2.2 Procedure

The Import page has four boxes: the **ACV standard routes**, the **ForeFlight NavLog**, the **Weather briefing** and the **Status**. The route comes either from a standard route or from a NavLog; the weather always from the briefing PDF.

1. **INPUT** part of a route name or of a waypoint in the search box (A) to narrow the list (optional).
2. **SELECT** the route card (B). The selected card is underlined; SELECT it again to clear it.
3. **VERIFY** the track (C) and the summary (D): number of waypoints, distance, Touch & Go aerodrome and the waypoints in order. **Show the waypoints** opens the waypoint table.

> **STANDARD ROUTES ACV**
> The Aeroclub Varese routes, read from the club's ForeFlight NavLogs: one card per route, with its track, distance and number of waypoints. The track is drawn from the magnetic tracks and distances of the NavLog, without wind; an out-and-back route is a single line. A route with a Touch & Go arrives in the wizard with the T&G aerodrome already ticked (Step 1). Loading a ForeFlight NavLog replaces the selected route.

Figure 2.1 – ACV Standard Routes



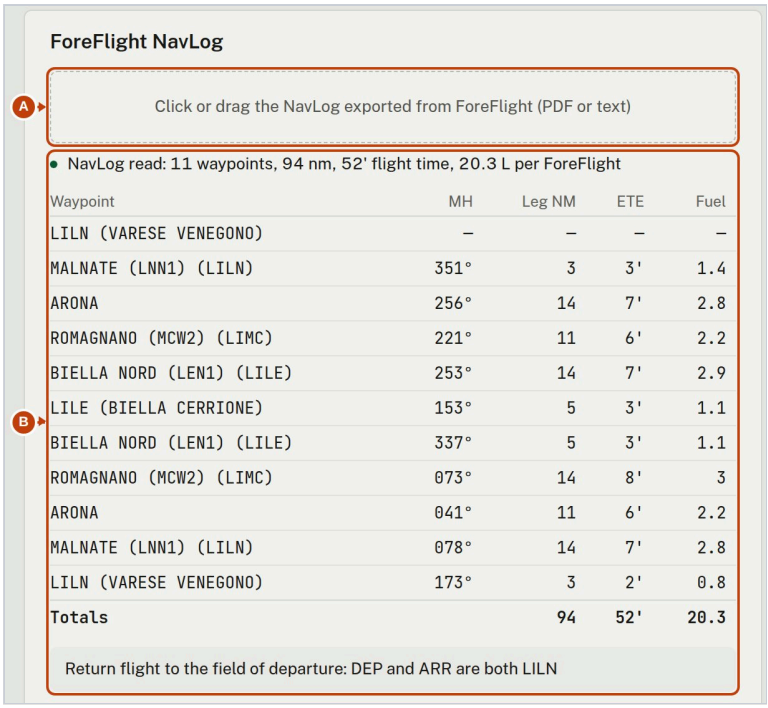
LILE Touch and Go selected.

A	**Search**	INPUT (optional)
B	**Route card**	SELECT
C	**Track**	VERIFY
D	**Route summary**	VERIFY

4. **LOAD** the ForeFlight NavLog in the NavLog drop zone (A), when no standard route is flown. Load it before the briefing: its ICAO codes are used to sort the briefing.
5. **VERIFY** the NavLog summary (B): number of waypoints, total distance and the waypoint table. For a return flight the application states that DEP and ARR are the same aerodrome.

> **NOTE**
> Only airports, route, magnetic tracks and distances are taken from the NavLog. ForeFlight speeds and fuel figures are not used: the AFM figures take precedence.

Figure 2.2 – NavLog Ingestion



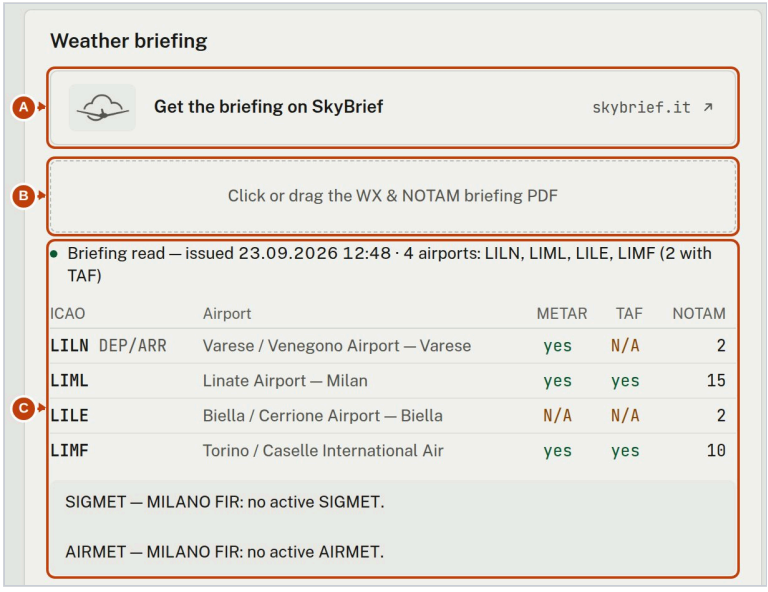
NavLog read: 11 waypoints, 94 NM, return flight to the departure aerodrome.

A	**NavLog drop zone**	LOAD the ForeFlight NavLog
B	**NavLog summary and waypoint table**	VERIFY waypoints and distance

6. **SELECT** **Get the briefing on SkyBrief** (A) if the briefing has not been downloaded yet: SkyBrief opens in a new tab. Search DEP, ARR and ALTN by ICAO and download the PDF.
7. **LOAD** the briefing PDF in the briefing drop zone (B).
8. **VERIFY** the briefing summary (C): issue date and time (UTC) and, for each aerodrome, the availability of METAR, TAF and the number of NOTAMs. `N/A` means not contained in the briefing.

> **NOTE**
> METAR, SPECI, TAF and NOTAM are sorted onto DEP, ARR, ALTN and the T&G aerodrome; from there wind, QNH and temperature feed the performance calculation (Step 4).

Figure 2.3 – Weather Briefing Ingestion



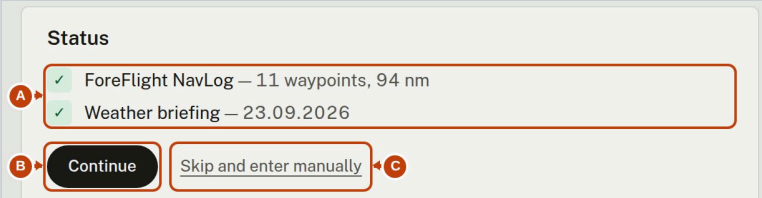
Four aerodromes found in the briefing, two of them with TAF.

A	**SkyBrief link**	SELECT to download the briefing
B	**Briefing drop zone**	LOAD the briefing PDF
C	**Briefing summary and aerodrome table**	VERIFY issue time and coverage

9. **CONFIRM** both STATUS items are green (A).
10. **SELECT** **Continue** (B) to open the wizard.

> **CAUTION**
> **Skip and enter manually** (C) opens the wizard without imported data. Route, distances and all weather and NOTAM fields must then be entered by hand. A missing or unreadable file does not block the wizard: its data are entered by hand.

Figure 2.4 – Import Status



Both files read: the wizard can be opened.

A	**Status**	CONFIRM both items green
B	**Continue**	SELECT to proceed
C	**Skip and enter manually**	Manual entry only

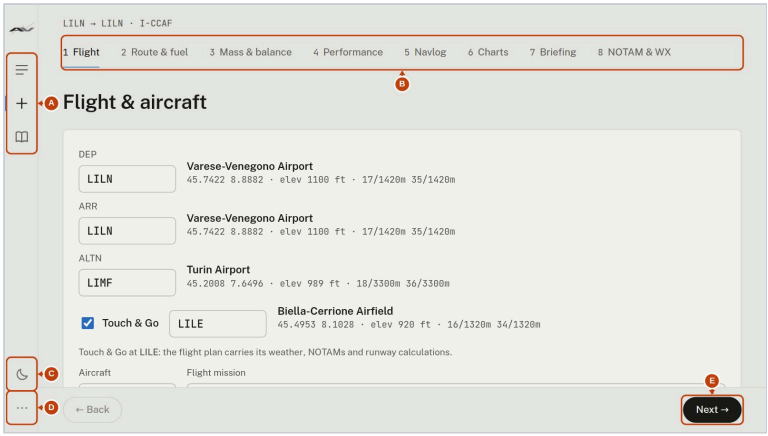
SECTION 3 Pre-flight Data Entry – Wizard

3.0 Wizard navigation

1. **SELECT** a step tab (B) to go to any step, or **Back** / **Next** (E) at the bottom of the page. On Step 8 **Next** becomes **Proceed** (Section 3.8).
2. **SELECT** in the side bar (A): **Flights** (the Dashboard, Section 4.5), **New flight** (the Import page) or **User manual** (this manual, over the page: **Close** or Esc goes back).
3. **SELECT** the theme button (C) to choose the display (below). It does not affect the documents.
4. **SELECT** the menu (D) for the document preview and the documents (Section 4.2).
5. **SELECT** **Save**, at the right end of the step tabs, to store the flight in your account at any step. The wizard stays where it is and the button reads **Saved** for a moment; a dot before **Save** means that there are changes not yet saved.

CAUTION
The underline marks the step in progress; steps already passed are shown dimmed. This does **not** mean that a step is complete or correct.

Figure 3.1 – Wizard Navigation



Side bar on the left, step tabs above the page.

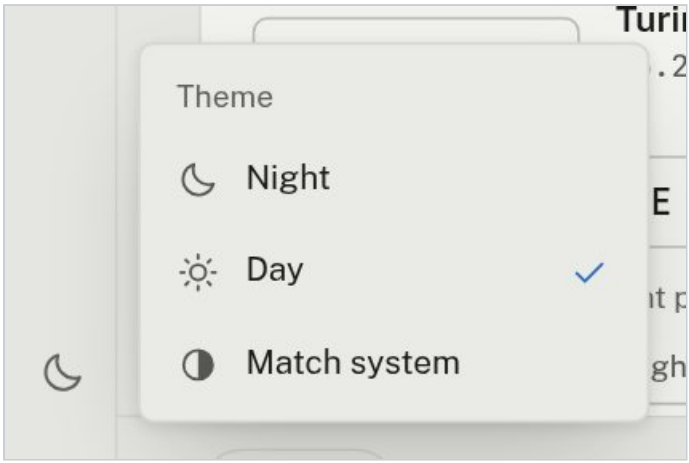
A	**Side bar**	Flights / New flight / User manual
B	**Step tabs 1–8**	Navigate between steps
C	**Theme**	Night / Day / Match system
D	**Menu**	Preview and documents
E	**Back / Next**	Previous / next step; Proceed on Step 8

NOTE
Leaving a flight with changes not saved (**Flights**, **New flight**, another flight or its documents, **Sign out**) asks first: **Save** saves and goes on, **Don't save** goes on without saving, **Cancel** stays. Closing or reloading the browser tab asks with the browser's own message.

With a mouse or a trackpad the side bar widens while the pointer is over it: the name of each item appears and the icons grow. The page underneath does not move. With touch only, the side bar stays narrow and its icons work directly.

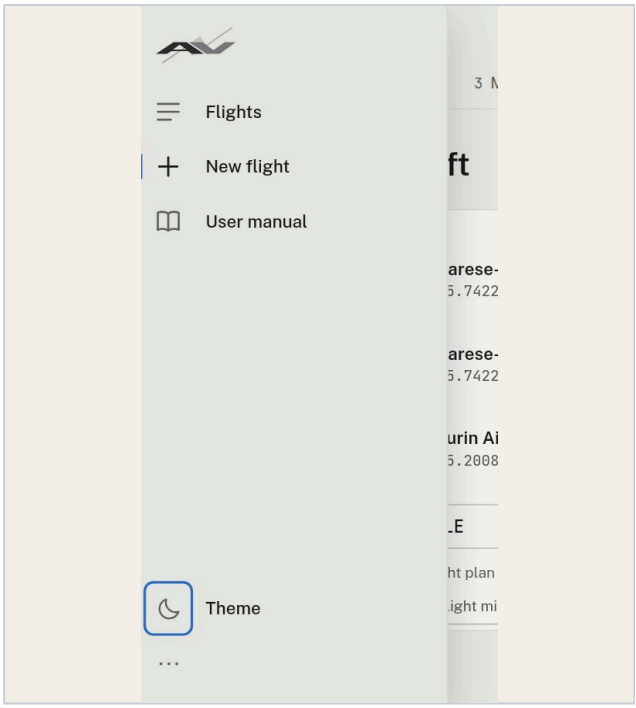
The theme button opens a choice of **Night** (dark), **Day** (light) and **Match system**, which follows the setting of the device. The new theme spreads from the button.

Figure 3.2 – Theme



The tick marks the current choice.

Figure 3.3 – Side Bar, Expanded



Shown while the pointer is over the side bar.

3.1 Step 1 – Flight & Aircraft

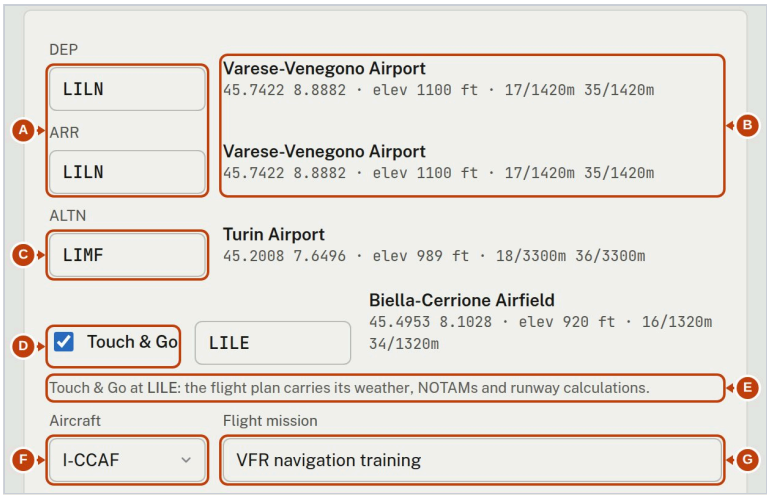
DATA ENTRY

FIELD	CLASS	SOURCE	PILOT ACTION
DEP / ARR	A	NavLog	VERIFY
Aerodrome data (name, position, elevation, runways)	A	OurAirports	VERIFY
ALTN	M	Pilot	INPUT
Touch & Go + ICAO	O	Proposed from route	SELECT if landing en route
Aircraft (registration)	M	I-CCAF default; Other = typed by the pilot	SELECT
Flight Mission	O	Pilot	INPUT
Route	A	NavLog	VERIFY
OFF, TO (UTC)	M	Pilot	SELECT
LDG, BLOCK, FLT	A	Computed	VERIFY
ON	A	Proposed, editable	VERIFY / CORRECT

1. **VERIFY** DEP and ARR (A) as imported from the NavLog.
2. **VERIFY** the aerodrome data line next to each code (B): name, coordinates, elevation and runways. The line appears automatically when the four-letter code is complete. No line = code not found: CROSS-CHECK the ICAO code.
3. **INPUT** the alternate ICAO code (C). Its weather, NOTAMs, distance and runway data are filled in automatically.
4. **SELECT** **Touch & Go** (D) only if a landing en route is planned. The application proposes the aerodromes found along the route (E).
5. **SELECT** the aircraft (F). Mass & Balance (Step 3) follows this choice. Select **Other** only for an aircraft not in the list or after a re-weighing: type its registration in the box that appears.
6. **INPUT** the flight mission (G) (optional).

> **NOTE**
> As soon as an ICAO code has four letters, its position, elevation and runways are looked up on OurAirports. From there come the alternate distance (ARR–ALTN), the elevations and the runways for Step 4. Every field remains editable by hand. The cruise altitude is set in Step 2.

Figure 3.4 – Aerodromes and Mission



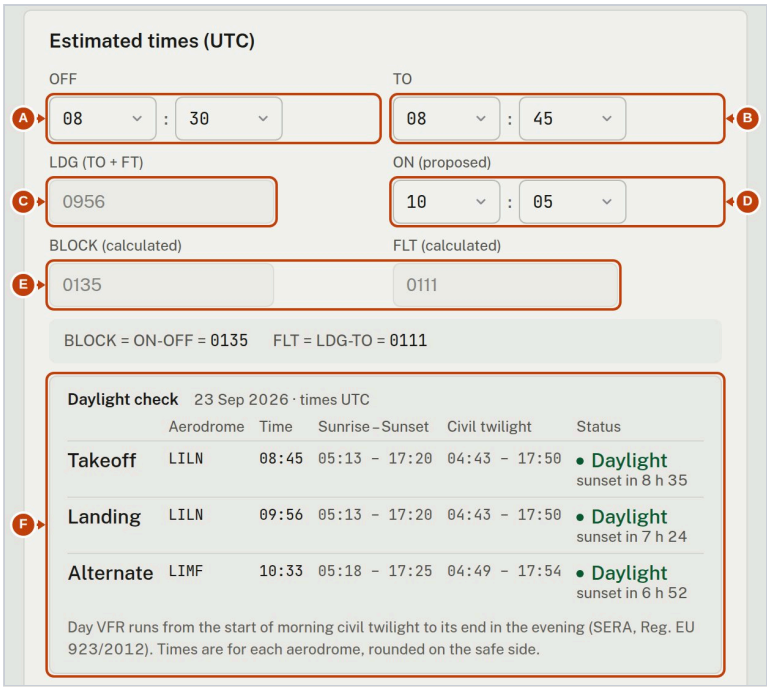
Aerodrome codes with the data looked up automatically.

A	**DEP / ARR**	VERIFY
B	**Aerodrome data line**	VERIFY elevation and runways
C	**ALTN**	INPUT
D	**Touch & Go**	SELECT if applicable
E	**Aerodromes along the route**	Information
F	**Aircraft**	SELECT
G	**Flight Mission**	INPUT (optional)

6. **SELECT** the off-blocks time OFF (A) and the take-off time TO (B), in UTC, 5-minute steps.
7. **VERIFY** LDG (C) = TO + Flight Time computed in Step 2.
8. **VERIFY** ON (D) = LDG + the departure taxi time. CORRECT it if required: once set by hand it no longer updates; set it back to `--:--` to restore automatic mode.
9. **VERIFY** BLOCK = ON − OFF and FLT = LDG − TO (E).
10. **REVIEW** the DAYLIGHT check (F). It requires the flight date (Step 8).

> **WARNING**
> Times are **UTC**. Times entered in local time make the daylight check wrong by the UTC offset.

Figure 3.5 – Estimated Times and Daylight Check



Take-off, landing at destination and landing at the alternate, each checked against its own aerodrome: sunrise – sunset, civil twilight and status.

A	**OFF**	SELECT (UTC)
B	**TO**	SELECT (UTC)
C	**LDG**	VERIFY
D	**ON**	VERIFY / CORRECT
E	**BLOCK / FLT**	VERIFY
F	**DAYLIGHT check**	REVIEW

DAYLIGHT CHECK – STATUS

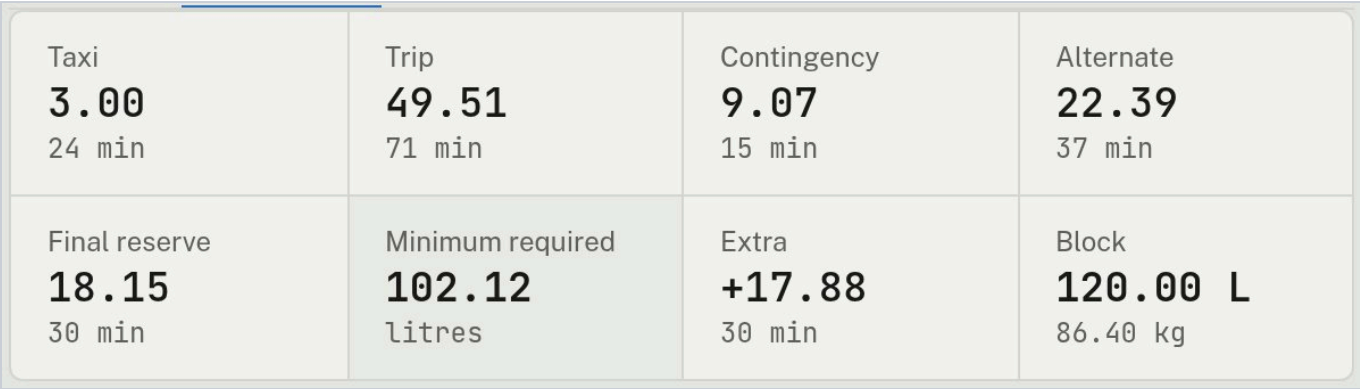
INDICATION	MEANING	ACTION
Daylight (green)	Between sunrise and sunset; the time left to sunset is shown	VERIFY margin
Civil twilight (amber)	After sunset or before sunrise, still day VFR; minutes past sunset and time left to night are shown	REVIEW plan
Night (red)	After the end / before the start of civil twilight	REVIEW plan
Not computed	Date, TO time, Trip time or coordinates missing; the box says which	INPUT missing data

The box border takes the colour of the worst row. On the OFP the same check is printed in compact form: `SS -h:mm`, `AFTER SUNSET (+h:mm)` / `BEFORE SUNRISE`, `NIGHT`, flagged `***`.

3.2 Step 2 – Route & Fuel

All fuel is in **litres**, fuel flows in **L/h**, speeds in **MPH**, as in the AFM; 1 L = 0.72 kg; 174 L (46 USG) usable. At the top of the step the **fuel plan** adds up Taxi, Trip, Contingency, Alternate, Final reserve, Minimum required, Extra and Block, as printed on the OFP: litres above, minutes below, Block also in kg. It updates with every value of this step.

Figure 3.6 – Fuel Plan



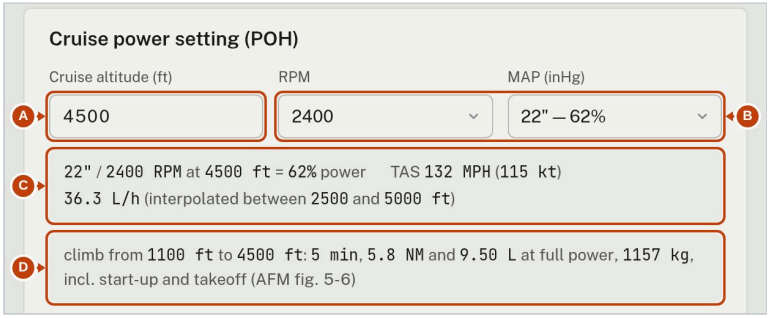
Block above 174 L is flagged on the Block box.

DATA ENTRY

FIELD	CLASS	SOURCE	PILOT ACTION
Cruise Altitude	M	Pilot	INPUT
RPM, MAP	M	AFM fig. 5-1 combinations only	SELECT
Trip distance	A	NavLog	VERIFY
Cruise GS	A	TAS of the selected setting (still air)	CORRECT for wind
Trip fuel flow	A	AFM fig. 5-1	VERIFY
Alternate distance	A	Great circle ARR–ALTN	VERIFY / CORRECT
Alternate altitude	M	Pilot	INPUT
Alternate GS, FF, Flight Time	A	22 inHg / 2400 RPM, locked	VERIFY
Flight Rules	M	VFR default	SELECT
Taxi time and fuel	A	Step 1 times, 7.5 L/h	VERIFY
Fuel On Board	O	Pilot (actual uplift)	INPUT
Extra (L)	O	Pilot, if FOB empty	INPUT

1. **INPUT** the cruise altitude (A). RPM and MAP stay greyed out until it is entered.
2. **SELECT** RPM and MAP (B). Only the combinations published by the AFM (fig. 5-1) at that altitude are offered: as the altitude increases, the maximum MAP drops by about one inch per 1000 ft and the higher-power combinations disappear. Selecting the setting fills in the Trip GS and fuel flow with the TAS and fuel flow of that row.
3. **VERIFY** the power line (C): % power, TAS and fuel flow.
4. **VERIFY** the climb line (D): from DEP elevation to cruise altitude, time, distance and fuel (start-up and take-off included).

Figure 3.7 – Cruise Power Setting



Selecting the setting fills in Trip GS and fuel flow.

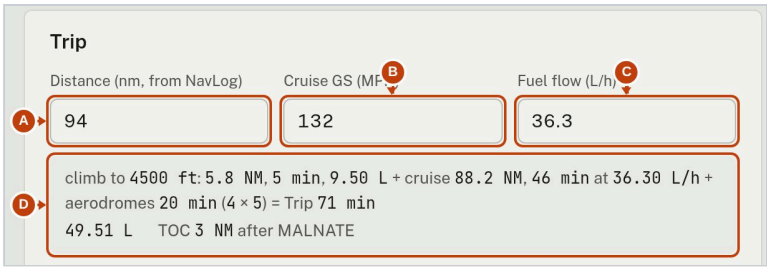
A	**Cruise Altitude**	INPUT
B	**RPM / MAP**	SELECT
C	**Power line**	VERIFY
D	**Climb line**	VERIFY

5. **VERIFY** the Trip distance **A** from the NavLog. It is the distance of the actual route, not the great circle DEP–ARR, and remains editable.
6. **CORRECT** the cruise GS **B** for the forecast wind. The proposed value is the TAS of the setting.
7. **VERIFY** the fuel flow **C**.
8. **VERIFY** the Trip line **D**: climb + cruise = Trip time and fuel, and the TOC position.

WARNING
An uncorrected GS assumes still air. With a headwind, Trip time and fuel are underestimated.

CAUTION
If the climb is longer than the trip, the line reads **Cruise altitude not reached**. REVIEW altitude or distance.

Figure 3.8 – Trip

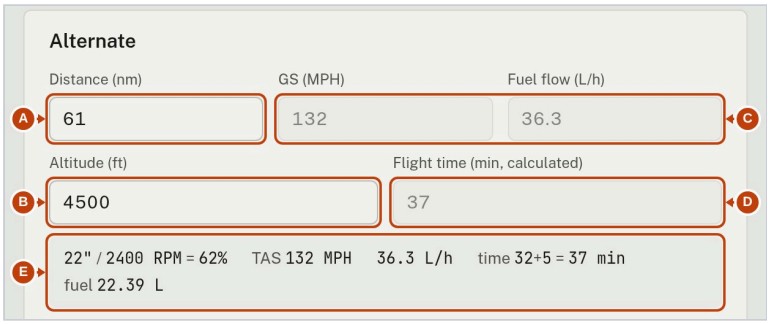


Trip = climb (AFM fig. 5-6) + cruise from the top of climb.

A	**Distance**	VERIFY
B	**Cruise GS**	CORRECT for wind
C	**Fuel flow**	VERIFY
D	**Trip line and TOC**	VERIFY

9. **VERIFY** the alternate distance **A** (great circle ARR–ALTN). CORRECT it for the planned routing.
10. **INPUT** the alternate altitude **B**.
11. **VERIFY** GS and fuel flow **C**, fixed at 22 inHg / 2400 RPM, and the computed Flight Time **D**.
12. **VERIFY** the alternate line **E**: setting, time and fuel.

Figure 3.9 – Alternate



The alternate always uses 22 inHg / 2400 RPM.

A	**Distance**	VERIFY / CORRECT
B	**Altitude**	INPUT
C	**GS / fuel flow (locked)**	VERIFY
D	**Flight Time (calculated)**	VERIFY
E	**Alternate line**	VERIFY

13. **SELECT** the flight rules (A): VFR = 30 min final reserve, IFR = 45 min.
14. **VERIFY** the reserves line (B): Contingency = max(5% of Trip, 15 min); Final Reserve at cruise fuel flow.
15. **VERIFY** the Plan Block against the 174 L usable (C).
16. **VERIFY** the taxi time and fuel (D), derived from OFF–TO and LDG–ON at 7.5 L/h.

> **CAUTION**
> **ABOVE the 174 L usable** on the Plan Block line: the plan exceeds tank capacity.

Figure 3.10 – Reserves & Margins



Reserves, tank capacity and taxi are computed, not entered.

A	**Flight Rules**	SELECT
B	**Contingency / Final Reserve**	VERIFY
C	**Plan Block vs usable fuel**	VERIFY
D	**Taxi**	VERIFY

17. **INPUT** the fuel on board after uplift (A) (optional). It is used only to compute the extra fuel and is not printed.
18. **INPUT** the extra fuel (B) by hand if FOB is left empty.
19. **VERIFY** the line (C): minimum required, extra litres and minutes.

> **WARNING**
> **FOB ... INSUFFICIENT** in red: the fuel on board is below the minimum required.

Figure 3.11 – Fuel On Board & Extra



With FOB entered, the extra is FOB minus the minimum required.

A	**Fuel On Board**	INPUT (optional)
B	**Extra (L)**	INPUT if FOB empty
C	**Minimum / extra line**	VERIFY

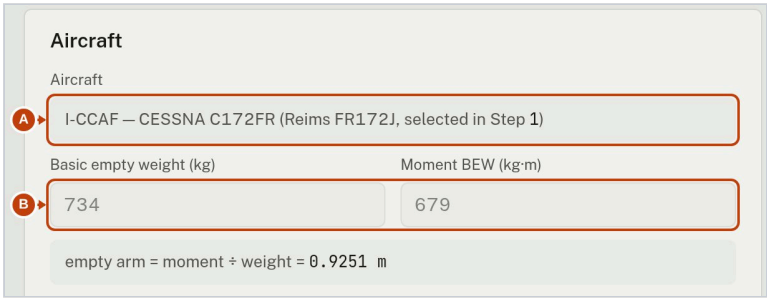
3.3 Step 3 – Mass & Balance

DATA ENTRY

FIELD	CLASS	SOURCE	PILOT ACTION
Aircraft	A	Selected in Step 1	VERIFY
Basic Empty Weight, Moment	A	Club W&B form; editable only with Other	VERIFY
Pilot + front pax	M	Pilot	INPUT
Rear pax	O	Pilot	INPUT if carried
Baggage (max 91 kg)	O	Pilot	INPUT if carried
EPOB / POB	M	Pilot	INPUT
Fuel masses	A	Step 2 (1 L = 0.72 kg)	VERIFY
MZFW, MTOW, MLAW	A	AFM limits	VERIFY
Remarks (M&B)	O	Pilot	INPUT

1. **VERIFY** the aircraft (A), selected in Step 1. Empty weight and moment (B) are filled in and locked. With **Other** (Step 1) they are entered by hand, for example after a re-weighing.

Figure 3.12 – Aircraft and Empty Weight



Empty weight and moment from the flying club form.

A	**Aircraft (from Step 1)**	VERIFY
B	**BEW / moment (locked)**	VERIFY

2. **INPUT** pilot and front passenger mass (A).
3. **INPUT** rear passengers (B) and baggage (C) if carried. Baggage limit 91 kg.
4. **INPUT** persons on board (D).
5. **VERIFY** the ZFW line (E).

Figure 3.13 – Loading



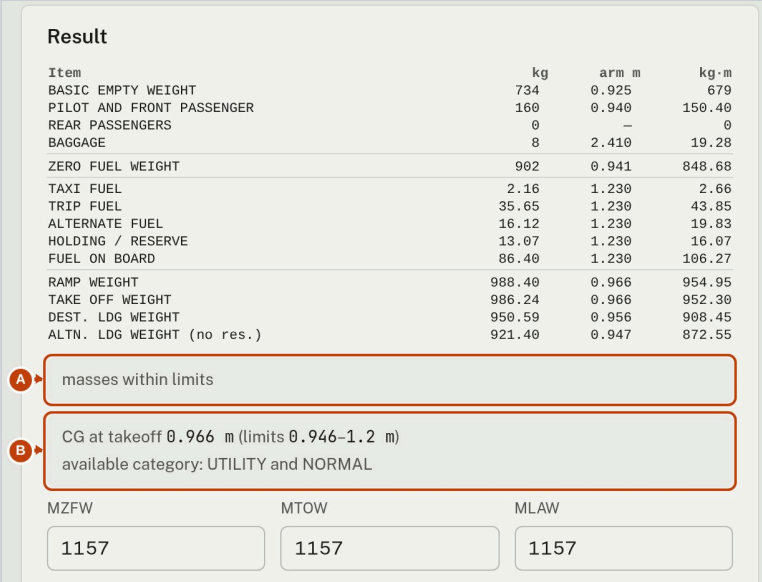
Fuel is not entered here: it comes from Step 2.

A	**Pilot + front pax**	INPUT
B	**Rear pax**	INPUT if carried
C	**Baggage**	INPUT if carried
D	**EPOB / POB**	INPUT
E	**ZFW line**	VERIFY

6. **VERIFY** the mass check (A): **masses within limits.**
7. **VERIFY** the CG check (B): CG at take-off within the envelope and the category available (UTILITY / NORMAL).
8. **VERIFY** the CG envelope chart: take-off, destination landing and alternate landing points inside the envelope.

> **WARNING**
> Any mass or CG message other than **within limits** must be resolved before flight.

Figure 3.14 – Mass & Balance Result



Loading sheet with the arms of the club form and the AFM limits.

A	**Mass check**	VERIFY
B	**CG and category**	VERIFY

3.4 Step 4 – Performance

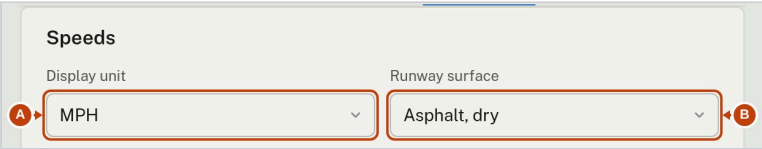
DATA ENTRY

FIELD	CLASS	SOURCE	PILOT ACTION
Display Unit	O	MPH default	SELECT (display only)
Runway Surface	M	Asphalt, dry default	SELECT
Runway (each aerodrome)	M	None pre-selected	SELECT
Elevation, RWY heading	A	OurAirports	VERIFY
TORA / TODA / LDA	A	OurAirports physical length	CROSS-CHECK with AIP
QNH, Temp, Wind	A	Own METAR (or nearest station)	VERIFY
Runway (OFP)	A	Runway selected	VERIFY

1. **SELECT** the display unit (A) (MPH or knots). Calculations do not change.
2. **SELECT** the runway surface (B): asphalt dry, grass dry, or wet / contaminated.

> **CAUTION**
> Wet / contaminated: the AFM publishes no data and no increment is applied. A red warning is shown.

Figure 3.15 – Speeds and Surface



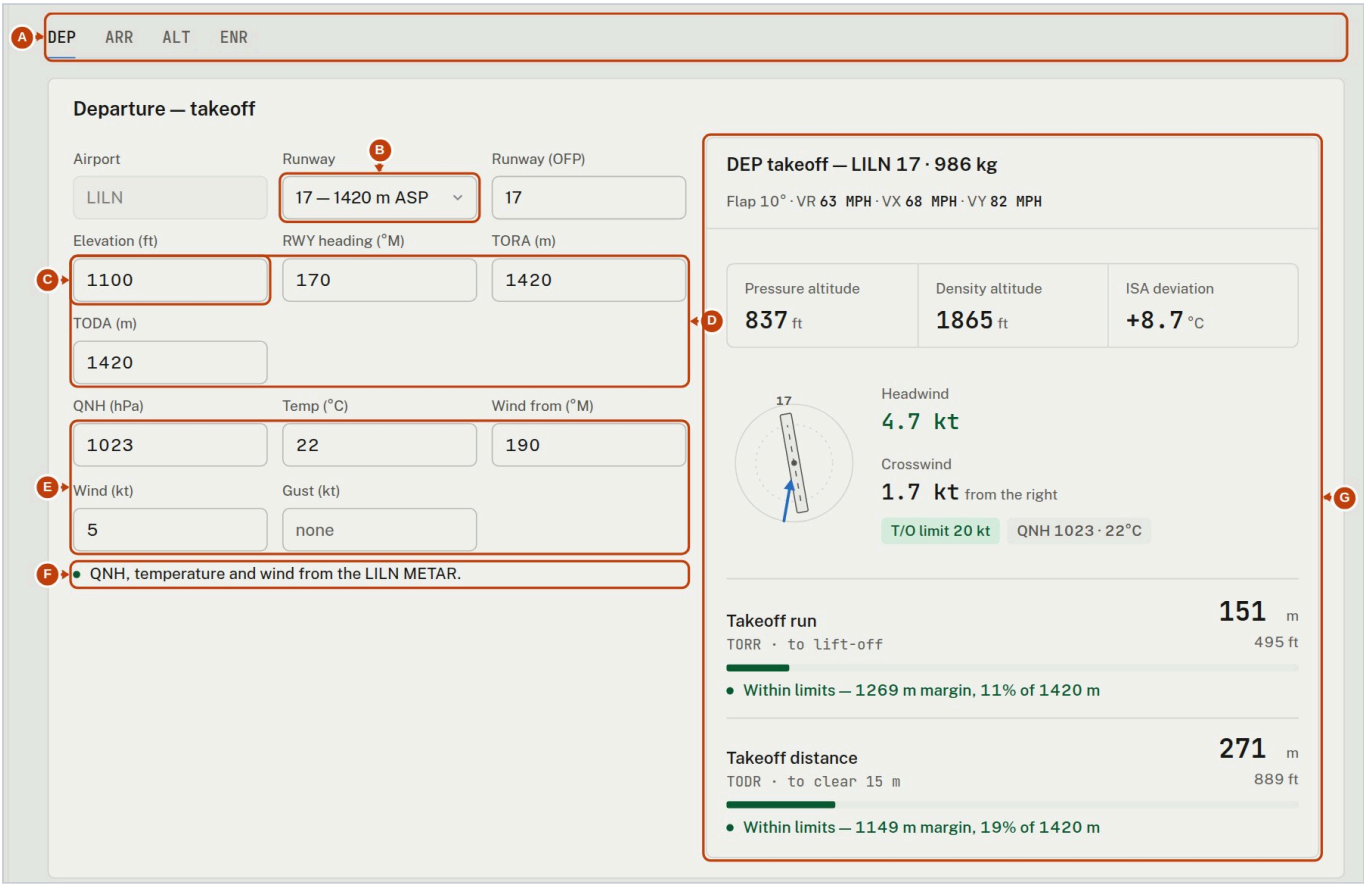
A	**Display Unit**	SELECT
B	**Runway Surface**	SELECT

Each phase has its own tab (A): **DEP** (take-off), **ARR** (landing), **ALT** (alternate landing) and **ENR** (touch and go, shown only when selected in Step 1). A dot on a tab indicates a warning for that phase.

3. **SELECT** each tab (A) in turn and carry out items 4 to 8.

4. **SELECT** the runway (B). No runway is pre-selected: select it from the wind of the day.
5. **VERIFY** the elevation (C) and the runway heading.
6. **CROSS-CHECK** TORA and TODA (LDA for landing) (D) with the AIP. They are pre-filled with the physical runway length.
7. **VERIFY** QNH, temperature, wind and gust (E), and their source (F). An empty **Wind From** means variable wind (VRB).
8. **VERIFY** the result card (G): pressure and density altitude, head/crosswind against the limit, distances required and margin.

Figure 3.16 – Take-off and Landing Data – DEP Tab



Inputs on the left, result card on the right. Wind, QNH and temperature come from the LILN METAR.

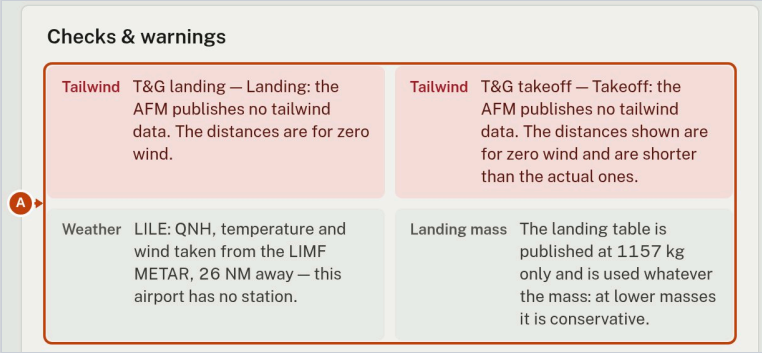
(A) **Phase tabs DEP / ARR / ALT / ENR**	SELECT each	(B) **Runway**	SELECT
(C) **Elevation**	VERIFY	(D) **TORA / TODA**	CROSS-CHECK with AIP
(E) **QNH / Temp / Wind**	VERIFY	(F) **Weather source**	VERIFY
(G) **Result card**	VERIFY margins		

9. **REVIEW** every message in **Checks & Warnings** (A). Red = the result does not cover the condition (e.g. tailwind, wet runway, standard weather used).

> **NOTE**
> An aerodrome without METAR uses standard values (1013 hPa, 15 °C, calm) and is marked in red. If a station with METAR exists within 30 NM, a button offers its values.

> **NOTE**
> Touch and go: landing and take-off are both computed with the take-off mass (conservative).

Figure 3.17 – Checks & Warnings



Example: tailwind on the T&G runway and weather taken from a nearby station.

A	**Checks & Warnings**	REVIEW every item

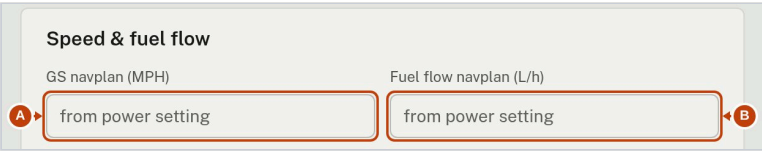
3.5 Step 5 – Navlog

DATA ENTRY

FIELD	CLASS	SOURCE	PILOT ACTION
GS / fuel flow navplan	O	Step 2 values if empty	INPUT only to override
Freq.	A	OurAirports (number only)	VERIFY
C/S	O	Pilot	INPUT
Checkpoint, MT, Dist.	A	NavLog	VERIFY
Alt.	A	Cruise altitude	VERIFY / CORRECT per waypoint (ft or FL)
Vertical profile	A	Computed from Alt.	VERIFY
EET, ETO, Fuel	A	Computed	VERIFY
Remarks	O	Pilot	INPUT

1. **VERIFY** GS and fuel flow (A) (B). Leave empty to use the Step 2 values. Times and fuel are not copied from ForeFlight: they are recomputed, so that the legs add up exactly to the Trip of the fuel plan.

Figure 3.18 – Navlog Speed & Fuel Flow

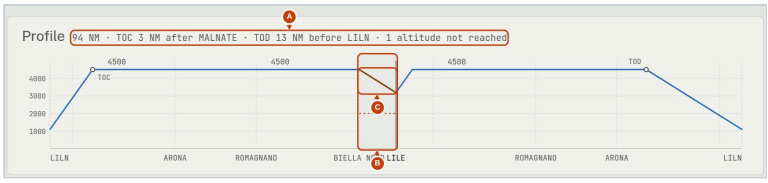


A	**GS navplan**	Leave empty or INPUT
B	**Fuel flow navplan**	Leave empty or INPUT

At the top of the step the **vertical profile** draws the flight to scale along the navplan route, leg by leg, with the altitudes of the **Alt.** column. It redraws while an altitude is typed. On a tablet or a computer it stays at the top of the page while the legs table is scrolled.

2. **VERIFY** the summary line (A): total distance, TOC position, TOD position and, if any, the number of altitudes not reached.
3. **VERIFY** the profile: climb, cruise altitudes, descents and the final descent to the arrival aerodrome. The leg of the row being edited is highlighted (B); SELECT a leg on the profile to go to its **Alt.** box.
4. **REVIEW** every leg drawn in amber (C): its altitude cannot be reached within the leg, and the wanted altitude is shown dashed. CORRECT the altitude or accept the leg as drawn.

Figure 3.19 – Vertical Profile



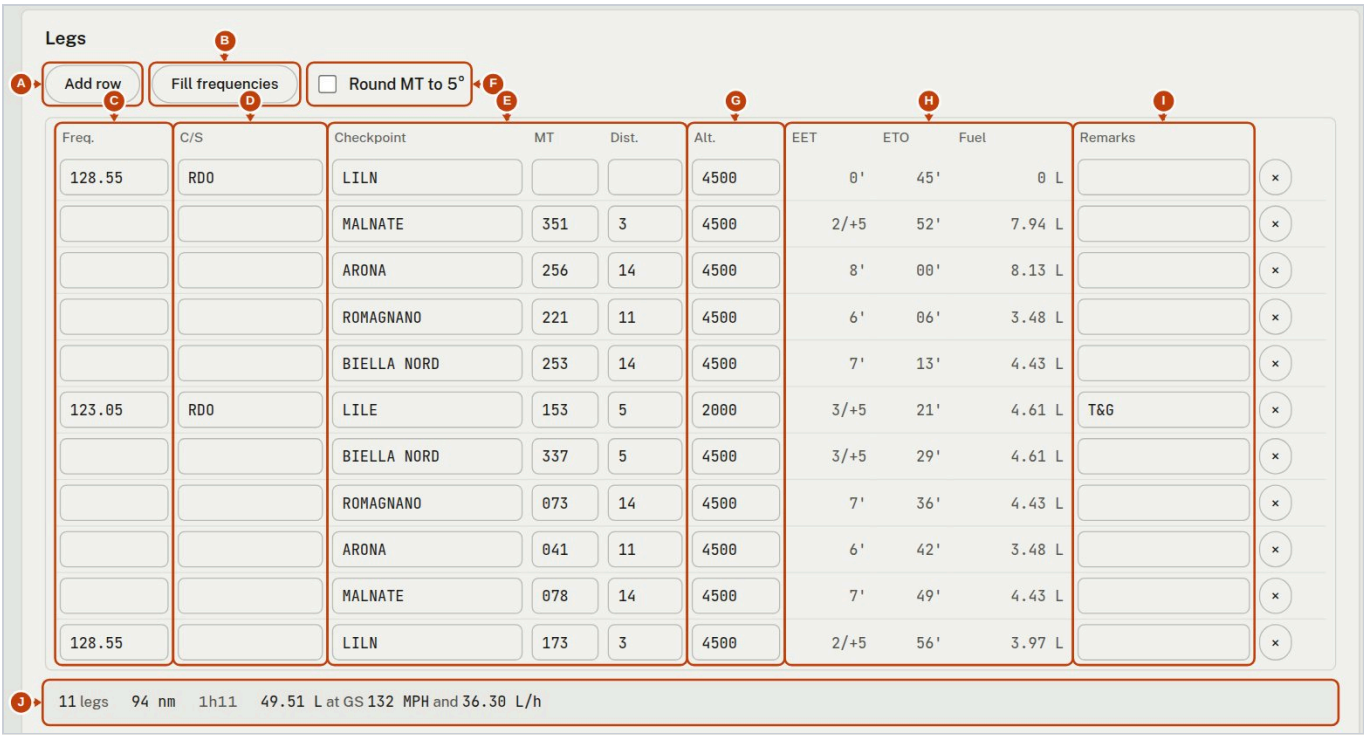
Example: the leg to LILE planned at 2000 ft cannot be reached from 4500 ft at 500 ft/min.

A	**Summary line**	VERIFY
B	**Leg being edited**	Information
C	**Altitude not reached**	REVIEW

> **NOTE**
> Climbs use the climb gradient of the fuel plan (AFM fig. 5-6, from the DEP elevation to the cruise altitude), so the TOC drawn is the TOC of the fuel plan. Descents are drawn at 500 ft/min at the navplan GS. The altitude of a row is the altitude of the leg that ends at that checkpoint; the change starts at the previous checkpoint. Descents are a drawing aid: times and fuel are not changed (Section 5.3).

5. **SELECT Add row** (A) to add a checkpoint, × to delete one.
6. **SELECT Fill frequencies** (B) to repeat the frequency look-up. VERIFY frequencies (C).
7. **INPUT** callsigns (D) (not in any database).
8. **VERIFY** checkpoints, magnetic tracks and distances (E).
9. **SELECT Round MT to 5°** (F) if required: magnetic tracks are shown and printed to the nearest 5° (347 → 345, 358 → 360). The ForeFlight values are kept; untick to restore them.
10. **VERIFY** the altitude of each waypoint (G), in feet or as a flight level (FL065); CORRECT where different from cruise. Changing the cruise altitude in Step 2 updates only the rows that still carry the old cruise altitude.
11. **VERIFY** EET per leg, ETO (minutes after TO) and fuel (H). A leg from or to an aerodrome reads e.g. `7/+5`: 7 min en route + 5 aerodrome minutes (Section 5.3), already included in ETO, Trip and fuel.
12. **INPUT** remarks (I) (optional).
13. **CROSS-CHECK** the totals line (J) with the Trip of Step 2: time and fuel must be the same.

Figure 3.20 – Navlog Legs



Times and fuel are recomputed from the selected power setting, not copied from ForeFlight; the climb is split over the legs by distance.

A **Add row**	Add checkpoint	**B** **Fill frequencies**	Repeat look-up
C **Freq.**	VERIFY	**D** **C/S**	INPUT
E **Checkpoint / MT / Dist.**	VERIFY	**F** **Round MT to 5°**	SELECT if required
G **Alt.**	VERIFY / CORRECT	**H** **EET / ETO / Fuel**	VERIFY
I **Remarks**	INPUT (optional)	**J** **Totals line**	CROSS-CHECK with Trip

> **NOTE**
> A leg longer than 10 minutes shows an advisory message (turning points ideally every 6–8 minutes). It does not block the wizard and does not change the navlog.

3.6 Step 6 – Destination Charts

1. **LOAD** up to 2 images or PDFs (aerodrome or approach charts) (optional).
2. **INPUT** the operational notes as required: Track, Runway, Altitudes, Clearance, Timing, On ground, Radios, Specials (optional).

Charts and notes are printed on the DEST CHARTS page of the OFP. Graphic pages of the briefing (e.g. significant weather, wind charts) are added to the OFP automatically, one per page.

3.7 Step 7 – Briefing

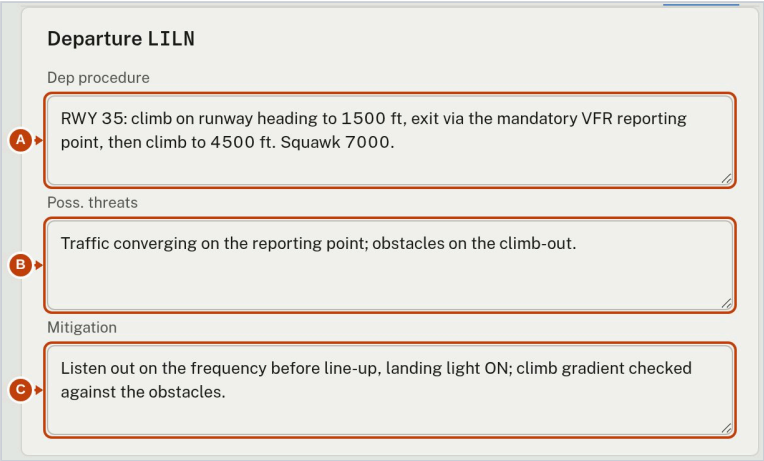
A briefing by phase of flight, written before departure, that gives a structure to everything that has to be done in flight. All fields are optional free text and nothing is checked by the application; they are printed on the BRIEFING page of the OFP (Section 4.3).

DATA ENTRY

FIELD	CLASS	SOURCE	PILOT ACTION
Procedure (DEP, ENR, T&G, DEST, ALTN)	O	Pilot	INPUT
Possible threats	O	Pilot	INPUT
Mitigation	O	Pilot	INPUT
Aircraft status	O	Pilot	INPUT if equipment is inoperative
Remarks	O	Pilot	INPUT if applicable

1. **INPUT** for the phase the **procedure** (A): what will be flown, e.g. runway, routing, reporting points, altitudes, frequencies.
2. **INPUT** the **possible threats** (B) of that phase, e.g. weather, terrain, traffic, airspace, aircraft, pilot.
3. **INPUT** the **mitigation** (C): what will be done to keep each threat under control.
4. **INPUT** the same three fields for Enroute, Touch & Go, Destination and Alternate. The Touch & Go block is shown only when Touch & Go is selected in Step 1; each heading shows the aerodrome code.

Figure 3.21 – Briefing – Departure



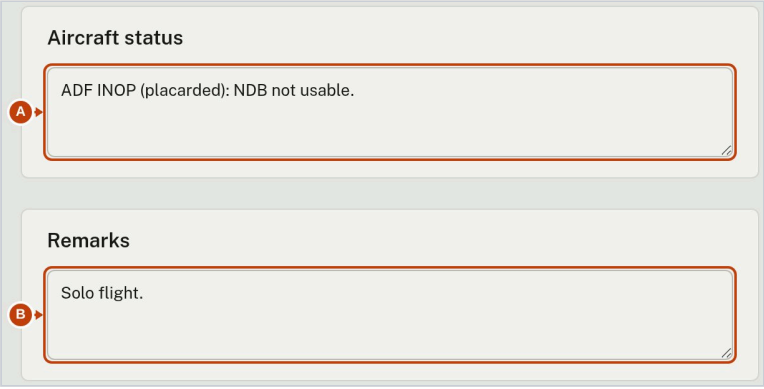
Example. Every phase has the same three fields.

A	**Procedure**	INPUT
B	**Possible threats**	INPUT
C	**Mitigation**	INPUT

1. **INPUT Aircraft status** (A) only when instruments or equipment are temporarily inoperative, stating which aids will therefore not be available in flight.
2. **INPUT Remarks** (B) only for important information on the conduct of the flight that is not related to the routing: exam flight, solo flight, dual command, SPIC flight.

> **NOTE**
> Keep the text short and in the order it will be used: the BRIEFING page is read in flight.

Figure 3.22 – Aircraft Status and Remarks



Example.

A	**Aircraft status**	INPUT if applicable
B	**Remarks**	INPUT if applicable

3.8 Step 8 – NOTAM & Weather

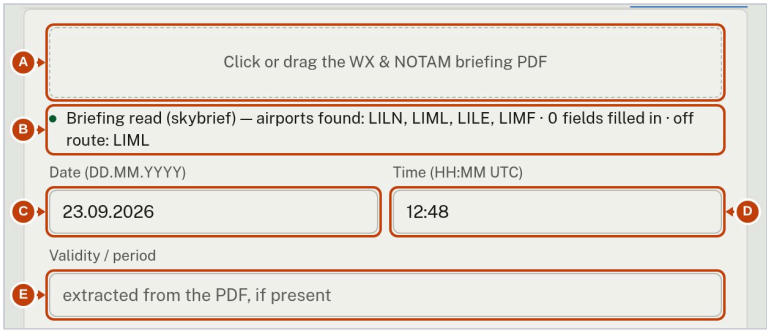
DATA ENTRY

FIELD	CLASS	SOURCE	PILOT ACTION
Briefing PDF	A	Loaded at import; can be reloaded	LOAD only to update
Date (DD.MM.YYYY)	M	Briefing	VERIFY / INPUT
Time (UTC), Validity	A	Briefing	VERIFY
METAR, SPECI, TAF, NOTAM per aerodrome	A	Briefing	VERIFY
Enroute NOTAMs, SIGMET / AIRMET	A	Briefing	VERIFY
PIC name	M	Pilot	INPUT

1. **LOAD** a newer briefing (A) only to update the weather.
2. **VERIFY** the status line (B): aerodromes found and those not on the route.
3. **VERIFY** the date (C). It is printed on the OFP and used by the daylight check. INPUT it if no briefing was loaded.
4. **VERIFY** issue time (D) and validity (E).

> **NOTE**
> If the briefing was issued on the day before the flight, the date shown is the issue date.

Figure 3.23 – Briefing Data



A	**Briefing drop zone**	LOAD to update
B	**Status line**	VERIFY
C	**Date**	VERIFY / INPUT
D	**Time (UTC)**	VERIFY
E	**Validity**	VERIFY

5. **VERIFY** for each aerodrome (DEP, ARR, T&G, ALTN) METAR (A), SPECI (B), TAF (C) and NOTAM (D): correct aerodrome, current issue.
6. **VERIFY** the Enroute NOTAM and SIGMET / AIRMET boxes. An absence must be stated (e.g. “no active SIGMET”): an empty box is not information.
7. **INPUT** the PIC name. It is printed on every page of the OFP and on the NAV-FLIGHTPLAN.
8. **SELECT** **Proceed** at the bottom of the page: the flight is saved to your account (READY TO FLY when the briefing is loaded) and the **Documents** page opens (Section 4.1).

> **NOTE**
> `N/A – not available in the briefing` is an explicit statement that the report is not contained in the briefing.

Figure 3.24 – Aerodrome Weather & NOTAM



One block per aerodrome. Text edited by hand is not overwritten by a later import.

A	**METAR**	VERIFY
B	**SPECI**	VERIFY
C	**TAF**	VERIFY
D	**NOTAM**	VERIFY

SECTION 4 System Outputs & Dispatch

4.1 Documents

The **Documents** page collects the three documents of a flight. It opens with **Proceed** on Step 8 of the wizard (Section 3.8), with **Documents** in the menu (Section 4.2) and with **Documents** on a READY TO FLY flight of the Dashboard (Section 4.5).

1. **VERIFY** the flight shown under the title: route, date and PIC.
2. **SELECT Download** (D) on the row of a document: the PDF is saved with the file name shown on the row (E).
3. **SELECT Download all** (C) to save the three PDFs one after the other.
4. **SELECT Edit flight** (B) to go back to the wizard, or **Flights** (A) to go back to the Dashboard.

DOCUMENT	FILE
Operational flight plan, A4 (Section 4.3)	`OFP_DEP-ARR_date.pdf`
NAV-FLIGHTPLAN, A5 (Section 4.4)	`NAVLOG_DEP-ARR_date.pdf`
Aeroclub Varese navlog, 2 × A4 (Section 4.4)	`CLUB-NAVLOG_DEP-ARR_date.pdf`

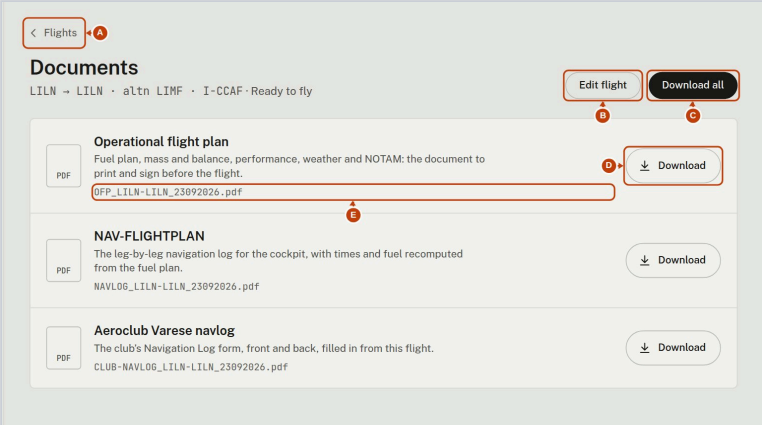
NOTE

The PDFs are generated from the flight data at the moment of the download: after a change in the wizard, download them again. The page has its own address (`#/flights/…/documents`): it can be reloaded or kept as a bookmark. On iPad, Safari may ask to allow multiple downloads the first time **Download all** is used. To print, use the print function of the PDF viewer.

CAUTION

Graphic weather charts are not stored with the flight: after reopening a saved flight they are in the OFP only if the briefing is loaded again. A flight without a briefing gives an OFP without METAR, TAF and NOTAM. The two navlogs need the legs of Step 5.

Figure 4.1 – Documents



Three documents of a READY TO FLY flight.

A	**Flights**	Back to the Dashboard
B	**Edit flight**	Back to the wizard
C	**Download all**	SELECT
D	**Download**	SELECT per document
E	**File name**	Information

4.2 Menu

ITEM	FUNCTION
Document Preview	Shows the OFP on screen
Documents	Saves the flight and opens its Documents page (Section 4.1)

The menu opens from **...** (More) at the bottom of the side bar. Below these items it shows the signed-in user, **Users** (administrator only), **Change password** and **Sign out**. Users and Change password open in a new tab. On the Dashboard the menu shows only the account items. The Dashboard, a new flight and this manual are in the side bar (Section 3.0); **Save** is at the right end of the step tabs.

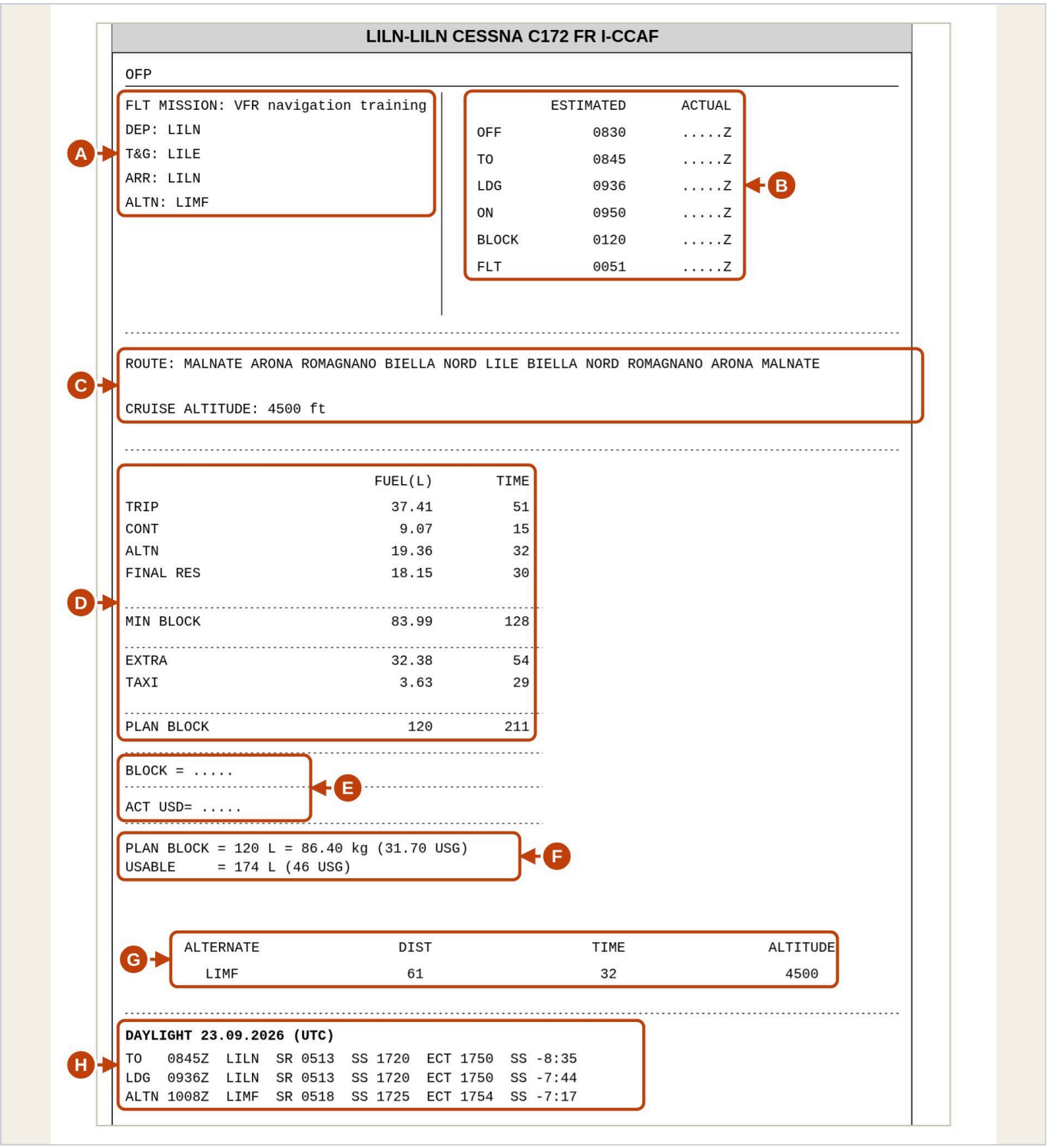
4.3 Operational Flight Plan (A4)

Downloaded from the Documents page (Section 4.1). **Document Preview** in the menu shows it on screen without downloading it.

OFP PAGE STRUCTURE (NUMBER OF PAGES VARIES WITH NOTAMS AND CHARTS)

PAGE	CONTENT
OFP	Flight data, estimated / actual times, route, cruise altitude, fuel plan, alternate summary, daylight check
MASS AND BALANCE	Loading sheet, limits, CG envelope chart, remarks, PIC signature
PERFORMANCE	Masses, V-speeds, take-off and landing distances and margins, cruise power setting, TOC, AFM cruise table
DEST CHARTS	Charts and operational notes (Step 6)
BRIEFING	Briefing by phase (DEP, ENR, T&G, DEST, ALTN): procedure, threats, mitigation; aircraft status, remarks (Step 7)
WEATHER	METAR / SPECI / TAF by aerodrome, SIGMET / AIRMET
NOTAM	NOTAM by aerodrome and enroute
Charts	Graphic pages of the briefing, if any

Figure 4.2 – OFP – Page 1



```
                    LILN-LILN CESSNA C172 FR I-CCAF

OFP

FLT MISSION: VFR navigation training
DEP: LILN
T&G: LILE
ARR: LILN
ALTN: LIMF

          ESTIMATED      ACTUAL
OFF           0830      .....Z
TO            0845      .....Z
LDG           0936      .....Z
ON            0950      .....Z
BLOCK         0120      .....Z
FLT           0051      .....Z

ROUTE: MALNATE ARONA ROMAGNANO BIELLA NORD LILE BIELLA NORD ROMAGNANO ARONA MALNATE

CRUISE ALTITUDE: 4500 ft

                          FUEL(L)        TIME
TRIP                        37.41          51
CONT                         9.07          15
ALTN                        19.36          32
FINAL RES                   18.15          30
MIN BLOCK                   83.99         128
EXTRA                       32.38          54
TAXI                         3.63          29
PLAN BLOCK                    120         211

BLOCK = .....
ACT USD= .....

PLAN BLOCK = 120 L = 86.40 kg (31.70 USG)
USABLE     = 174 L (46 USG)

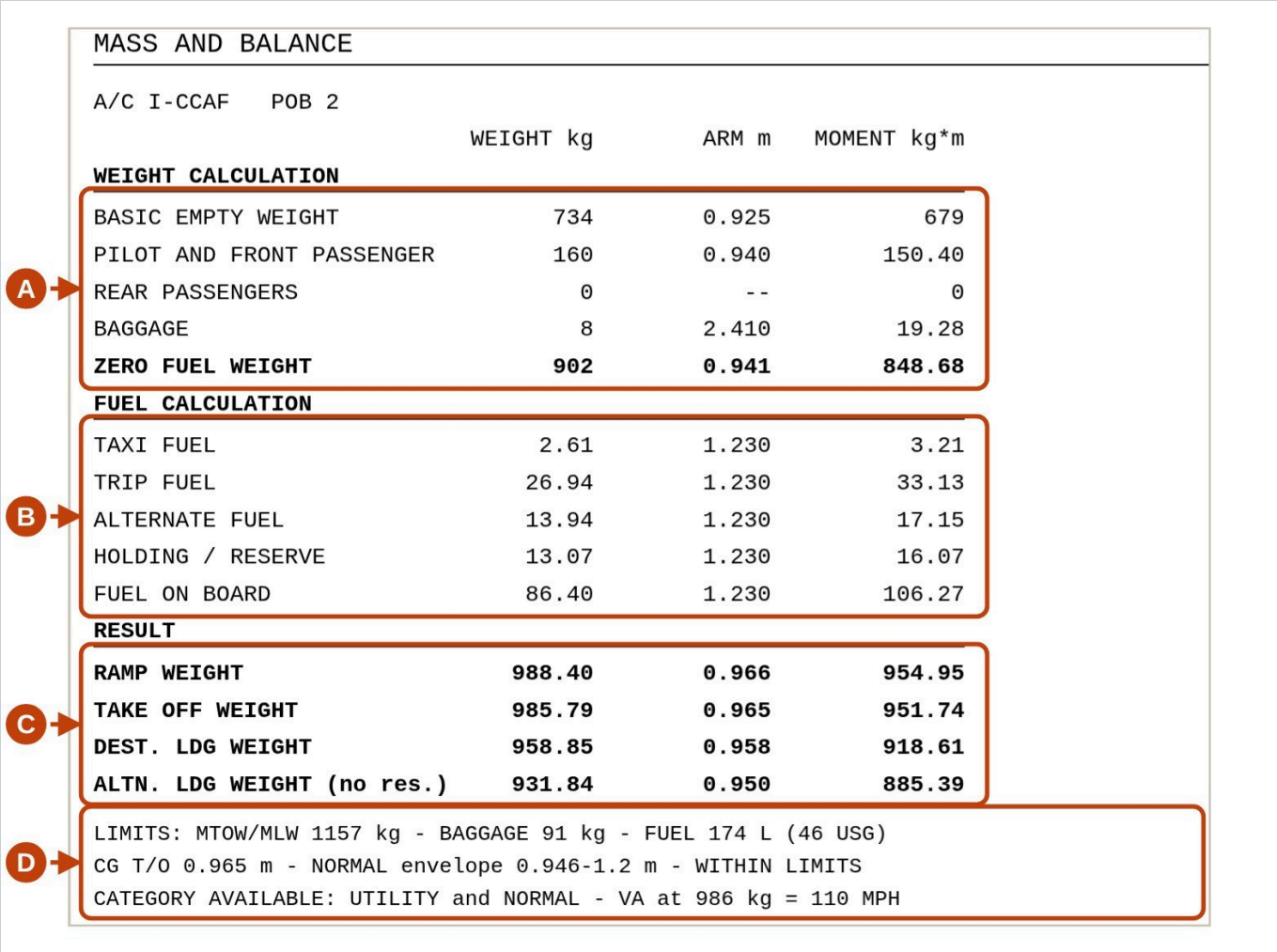
ALTERNATE                DIST                TIME                ALTITUDE
   LIMF                    61                  32                    4500

DAYLIGHT 23.09.2026 (UTC)
TO   0845Z  LILN  SR 0513  SS 1720  ECT 1750  SS -8:35
LDG  0936Z  LILN  SR 0513  SS 1720  ECT 1750  SS -7:44
ALTN 1008Z  LIMF  SR 0518  SS 1725  ECT 1754  SS -7:17
```

The ACTUAL column and the BLOCK / ACT USD lines are completed by hand.

A **Flight data**	DEP / T&G / ARR / ALTN	**B** **Estimated / actual times**	Record ACTUAL in flight (UTC)
C **Route and cruise altitude**	VERIFY	**D** **Fuel plan**	CROSS-CHECK with Step 2
E **BLOCK / ACT USD**	Complete by hand	**F** **Plan block mass, usable fuel**	VERIFY
G **Alternate summary**	VERIFY	**H** **Daylight check**	REVIEW

Figure 4.3 – OFP – Mass and Balance



MASS AND BALANCE

A/C I-CCAF POB 2

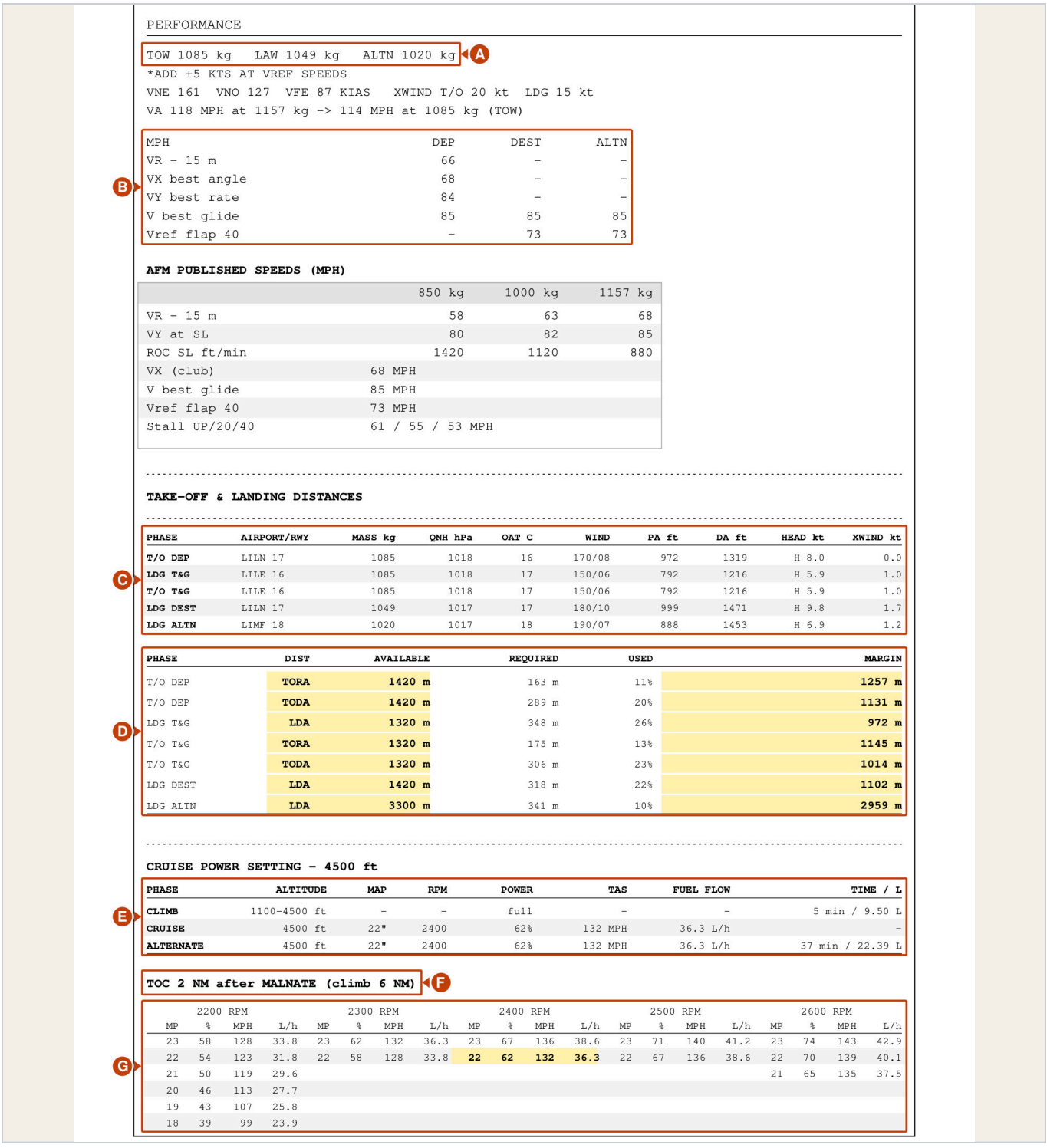
	WEIGHT kg	ARM m	MOMENT kg*m
WEIGHT CALCULATION			
BASIC EMPTY WEIGHT	734	0.925	679
PILOT AND FRONT PASSENGER	160	0.940	150.40
REAR PASSENGERS	0	--	0
BAGGAGE	8	2.410	19.28
ZERO FUEL WEIGHT	**902**	**0.941**	**848.68**
FUEL CALCULATION			
TAXI FUEL	2.61	1.230	3.21
TRIP FUEL	26.94	1.230	33.13
ALTERNATE FUEL	13.94	1.230	17.15
HOLDING / RESERVE	13.07	1.230	16.07
FUEL ON BOARD	86.40	1.230	106.27
RESULT			
RAMP WEIGHT	**988.40**	**0.966**	**954.95**
TAKE OFF WEIGHT	**985.79**	**0.965**	**951.74**
DEST. LDG WEIGHT	**958.85**	**0.958**	**918.61**
ALTN. LDG WEIGHT (no res.)	**931.84**	**0.950**	**885.39**

LIMITS: MTOW/MLW 1157 kg - BAGGAGE 91 kg - FUEL 174 L (46 USG)
CG T/O 0.965 m - NORMAL envelope 0.946-1.2 m - WITHIN LIMITS
CATEGORY AVAILABLE: UTILITY and NORMAL - VA at 986 kg = 110 MPH

Followed on the same page by the CG envelope chart, remarks and PIC signature.

A **Weight calculation**	VERIFY	B **Fuel calculation**	VERIFY
C **Result**	VERIFY	D **Limits and CG**	VERIFY within limits

Figure 4.4 – OFP – Performance



PERFORMANCE

TOW 1085 kg LAW 1049 kg ALTN 1020 kg (A)
*ADD +5 KTS AT VREF SPEEDS
VNE 161 VNO 127 VFE 87 KIAS XWIND T/O 20 kt LDG 15 kt
VA 118 MPH at 1157 kg -> 114 MPH at 1085 kg (TOW)

MPH	DEP	DEST	ALTN
VR – 15 m	66	–	–
VX best angle	68	–	–
VY best rate	84	–	–
V best glide	85	85	85
Vref flap 40	–	73	73

(B)

AFM PUBLISHED SPEEDS (MPH)

	850 kg	1000 kg	1157 kg
VR – 15 m	58	63	68
VY at SL	80	82	85
ROC SL ft/min	1420	1120	880
VX (club)	68 MPH		
V best glide	85 MPH		
Vref flap 40	73 MPH		
Stall UP/20/40	61 / 55 / 53 MPH		

TAKE-OFF & LANDING DISTANCES

PHASE	AIRPORT/RWY	MASS kg	QNH hPa	OAT C	WIND	PA ft	DA ft	HEAD kt	XWIND kt
T/O DEP	LILN 17	1085	1018	16	170/08	972	1319	H 8.0	0.0
LDG T&G	LILE 16	1085	1018	17	150/06	792	1216	H 5.9	1.0
T/O T&G	LILE 16	1085	1018	17	150/06	792	1216	H 5.9	1.0
LDG DEST	LILN 17	1049	1017	17	180/10	999	1471	H 9.8	1.7
LDG ALTN	LIMF 18	1020	1017	18	190/07	888	1453	H 6.9	1.2

(C)

PHASE	DIST	AVAILABLE	REQUIRED	USED	MARGIN
T/O DEP	TORA	1420 m	163 m	11%	1257 m
T/O DEP	TODA	1420 m	289 m	20%	1131 m
LDG T&G	LDA	1320 m	348 m	26%	972 m
T/O T&G	TORA	1320 m	175 m	13%	1145 m
T/O T&G	TODA	1320 m	306 m	23%	1014 m
LDG DEST	LDA	1420 m	318 m	22%	1102 m
LDG ALTN	LDA	3300 m	341 m	10%	2959 m

(D)

CRUISE POWER SETTING – 4500 ft

PHASE	ALTITUDE	MAP	RPM	POWER	TAS	FUEL FLOW	TIME / L
CLIMB	1100-4500 ft	–	–	full	–	–	5 min / 9.50 L
CRUISE	4500 ft	22"	2400	62%	132 MPH	36.3 L/h	–
ALTERNATE	4500 ft	22"	2400	62%	132 MPH	36.3 L/h	37 min / 22.39 L

(E)

TOC 2 NM after MALNATE (climb 6 NM) (F)

2200 RPM				2300 RPM				2400 RPM				2500 RPM				2600 RPM			
MP	%	MPH	L/h	MP	%	MPH	L/h	MP	%	MPH	L/h	MP	%	MPH	L/h	MP	%	MPH	L/h
23	58	128	33.8	23	62	132	36.3	23	67	136	38.6	23	71	140	41.2	23	74	143	42.9
22	54	123	31.8	22	58	128	33.8	**22**	**62**	**132**	**36.3**	22	67	136	38.6	22	70	139	40.1
21	50	119	29.6													21	65	135	37.5
20	46	113	27.7																
19	43	107	25.8																
18	39	99	23.9																

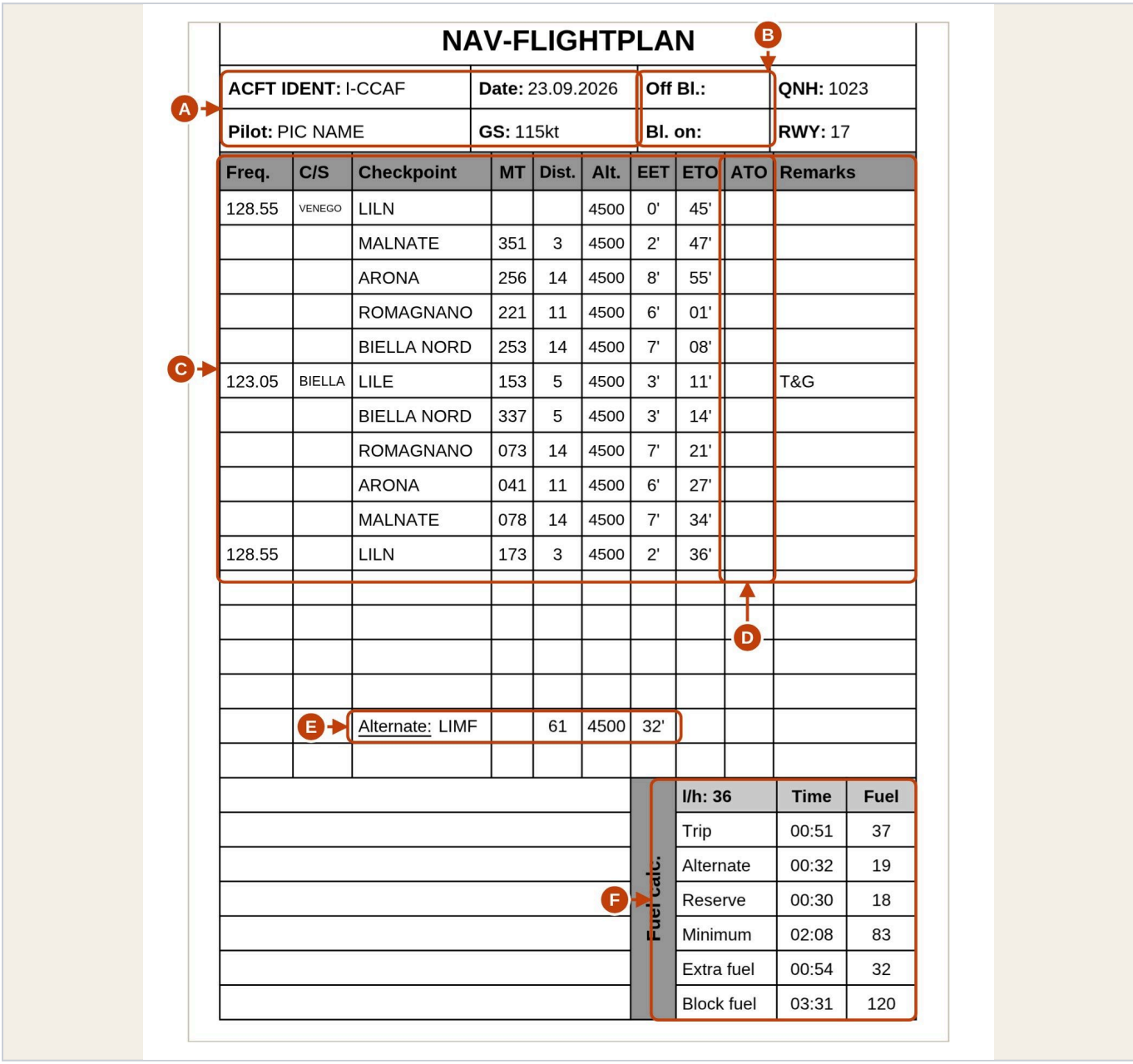
(G)

(A) Masses (TOW / LAW / ALTN)	VERIFY	(B) V-speeds by aerodrome	VERIFY
(C) Conditions by phase	VERIFY	(D) Distances available / required / margin	CROSS-CHECK
(E) Cruise power setting	VERIFY	(F) TOC position	VERIFY
(G) AFM cruise table	Reference		

4.4 NAV-FLIGHTPLAN (A5) and club navlog

1. **SELECT** **Download** on the NAV-FLIGHTPLAN row of the Documents page. It is a separate document and never part of the OFP.
2. **VERIFY** the header (A): aircraft, pilot, date, GS, QNH and runway.
3. **RECORD** off-blocks and on-blocks times (B) by hand.
4. **VERIFY** the legs (C).
5. **RECORD** ATO (D) in flight at each checkpoint.
6. **VERIFY** the alternate row (E) and the fuel calculation box (F).

Figure 4.5 – NAV-FLIGHTPLAN



NAV-FLIGHTPLAN

ACFT IDENT: I-CCAF	**Date:** 23.09.2026	**Off Bl.:**	**QNH:** 1023
Pilot: PIC NAME	**GS:** 115kt	**Bl. on:**	**RWY:** 17

Freq.	C/S	Checkpoint	MT	Dist.	Alt.	EET	ETO	ATO	Remarks
128.55	VENEGO	LILN			4500	0'	45'		
		MALNATE	351	3	4500	2'	47'		
		ARONA	256	14	4500	8'	55'		
		ROMAGNANO	221	11	4500	6'	01'		
		BIELLA NORD	253	14	4500	7'	08'		
123.05	BIELLA	LILE	153	5	4500	3'	11'		T&G
		BIELLA NORD	337	5	4500	3'	14'		
		ROMAGNANO	073	14	4500	7'	21'		
		ARONA	041	11	4500	6'	27'		
		MALNATE	078	14	4500	7'	34'		
128.55		LILN	173	3	4500	2'	36'		
		Alternate: LIMF		61	4500	32'			

Fuel calc.

l/h: 36	Time	Fuel
Trip	00:51	37
Alternate	00:32	19
Reserve	00:30	18
Minimum	02:08	83
Extra fuel	00:54	32
Block fuel	03:31	120

A5 document for the cockpit.

(A) **Header**	VERIFY	(B) **Off Bl. / Bl. on**	Record by hand
(C) **Legs**	VERIFY	(D) **ATO**	Record in flight
(E) **Alternate row**	VERIFY	(F) **Fuel calculation**	VERIFY

The **Aeroclub Varese navlog** produces the same navigation log on the flying club form (Aeroclub di Varese, Navigation Log): one PDF of two A4 landscape pages, front and back, to be printed double-sided. It does not replace the NAV-FLIGHTPLAN, which remains the navigation log of the application.

1. **SELECT** **Download** on the Aeroclub Varese navlog row of the Documents page.
2. **VERIFY** the main table (right side of the front): checkpoints, NAV AIDS (Ident = callsign, Freq), MC, ALT, DIST and FUEL per leg and remaining, ETE and ETO, as in the NAV-FLIGHTPLAN. Up to 13 checkpoints fit; up to 6 more continue in the left table, above the Dest. Airport row.
3. **VERIFY** the alternate (left side of the front): one leg from destination to alternate. The MC is computed from the great circle and the WMM2025 magnetic variation, and is marked as to be verified.
4. **VERIFY** the back: weather by aerodrome from the METAR; weight and balance with the take-off fuel; fuel calculation; take-off data (DEP) and landing data (DEST).
5. **RECORD** block times, RETO and ATO by hand.

> **NOTE**
> Differences from the club form as printed by hand: CLIMB fuel is the AFM fig. 5-6 block (start-up and take-off included), so no rate is given; DESC. is not published in the AFM; APP. is the 5 minutes per aerodrome; the contingency, which the form does not have, is included in MINIMUM FOB and stated in the Note. ASDA (required) is the take-off ground roll plus the landing ground roll. Emax is the best glide speed. ASDA declared, VLO and VLE are left blank; VX is the figure given by the club for each aircraft. All values are whole numbers: distances, leg fuel and masses to the nearest unit; in the fuel calculation the required quantities are rounded up and each total is the sum of the figures printed, so the columns add up. Arms stay in metres with three decimals.

> **CAUTION**
> The form is an aid to flight preparation and does not replace the AFM, the SOP or the AIP. Every pre-filled value must be verified by the pilot before use.

Figure 4.6 – Club Navlog – Front

Date: 08.10.2026	Dest. LILN	Altn. LIMF	
Route/Note LILN DCT LIMF			
MC 235: great circle 239T, var 4E (WMM2025) - to be verified			

Check Point	NAV AIDS	ROUTE	ALT	DIST	FUEL	Landing		Block on	
Dest. Airport	Ident	MC	Ft	leg	leg	ETE	ETO	RETO	ATO
LILN 128.55	Freq			rem.	rem.				
LIMF		235	4500	61	23	32/+5			
	118.50			0	64				
Alternate Airport	TOTAL								
LIMF 118.50				61	23	37'			

PRE-FLIGHT CHECKLIST	
Notams	Check
Wheather	Check
Charts	Organized
Flight log	Filled
A/m Documents	Check
Crew Documents	Check
Equipment	Check
Fuel	Check
Weight and Balance	Check
Flight Plan	Filed
Custom	Executed
Pre-flight inspection	Completed

POST FLIGHT CHECKLIST	
Flight Plan	Closed
Post-flight insp.	Completed
Handling duties	Executed
Custom	Executed

Aeroclub di Varese NAVIGATION LOG

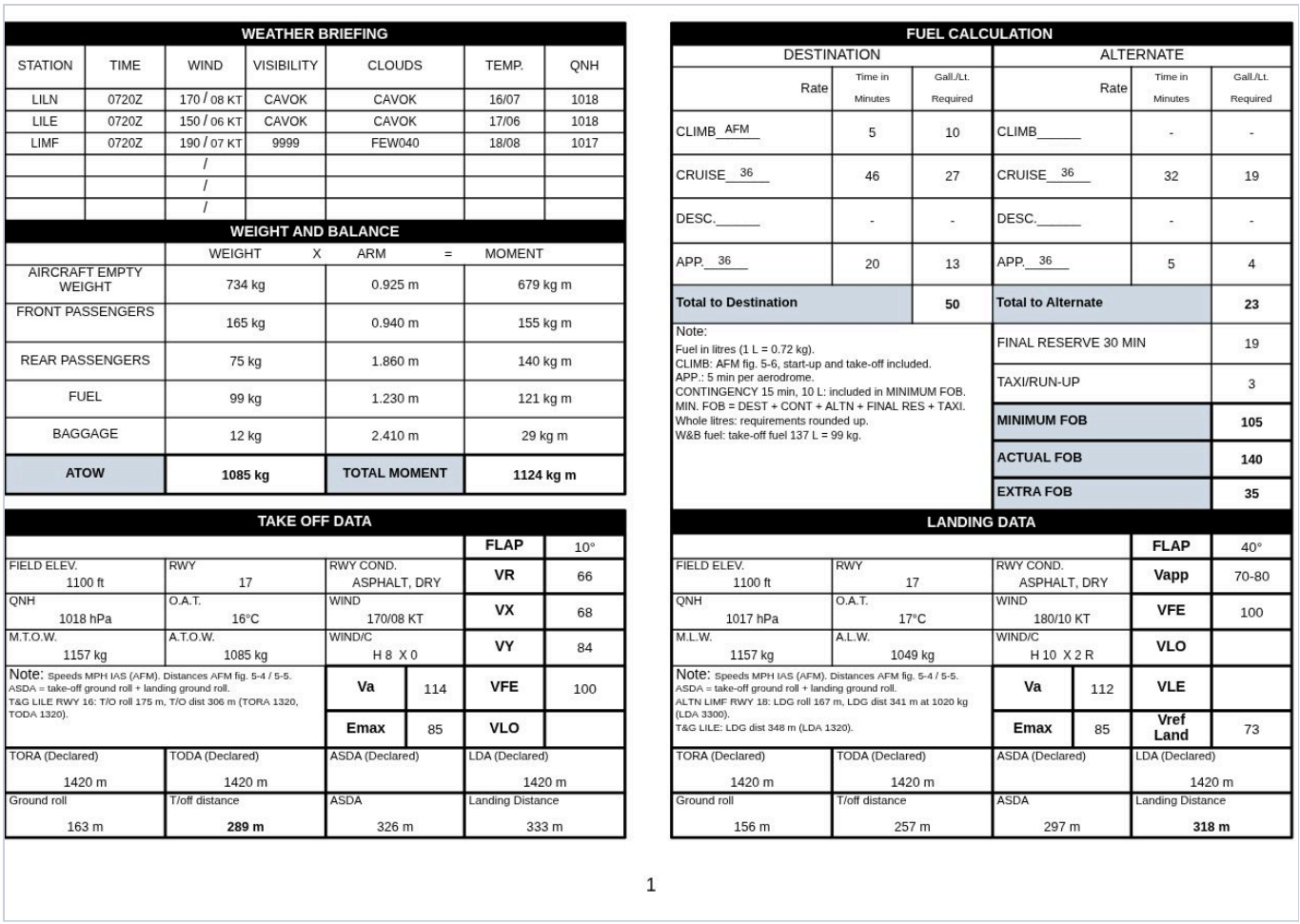
Date: 08.10.2026	Dept. LILN	Dest. LILN	Aircraft: I-CCAF FR172J
Route/Note MALNATE ARONA ROMAGNANO BIELLA NORD LILE BIELLA NORD ROMAGNANO ARONA MALNATE			

Check Point	NAV AIDS	ROUTE	ALT	DIST	FUEL	Block off / Landing		Take-off / Block on	
Dep. Airport	Ident	MC	Ft	leg	leg	ETE	ETO	RETO	ATO
LILN 128.55	Freq			rem.	rem.				
MALNATE		347	4500	4	9	3/+5	18'		
				90	128				
ARONA		268	4500	14	7	8'	26'		
				76	121				
ROMAGNANO		222	4500	9	3	5'	31'		
				67	118				
BIELLA NORD		268	4500	13	4	6'	37'		
				54	114				
LILE		181	4500	7	6	4/+5	46'		
	123.05			47	108				
BIELLA NORD		001	4500	7	5	4/+5	55'		
				40	103				
ROMAGNANO		088	4500	12	4	6'	01'		
				28	99				
ARONA		042	4500	10	3	5'	06'		
				18	96				
MALNATE		088	4500	14	4	7'	14'		
				4	92				
LILN		167	4500	4	5	2/+5	21'		
	128.55			0	87				
Destination Airport	TOTAL								
LILN 128.55				94	50	71'			

1

Main table on the right, alternate on the left.

Figure 4.7 – Club Navlog – Back



WEATHER BRIEFING

STATION	TIME	WIND	VISIBILITY	CLOUDS	TEMP.	QNH
LILN	0720Z	170 / 08 KT	CAVOK	CAVOK	16/07	1018
LILE	0720Z	150 / 06 KT	CAVOK	CAVOK	17/06	1018
LIMF	0720Z	190 / 07 KT	9999	FEW040	18/08	1017
		/				
		/				
		/				

WEIGHT AND BALANCE

	WEIGHT	X	ARM	=	MOMENT
AIRCRAFT EMPTY WEIGHT	734 kg		0.925 m		679 kg m
FRONT PASSENGERS	165 kg		0.940 m		155 kg m
REAR PASSENGERS	75 kg		1.860 m		140 kg m
FUEL	99 kg		1.230 m		121 kg m
BAGGAGE	12 kg		2.410 m		29 kg m
ATOW	**1085 kg**		**TOTAL MOMENT**		**1124 kg m**

TAKE OFF DATA

				FLAP	10°
FIELD ELEV. 1100 ft	RWY 17	RWY COND. ASPHALT, DRY		**VR**	66
QNH 1018 hPa	O.A.T. 16°C	WIND 170/08 KT		**VX**	68
M.T.O.W. 1157 kg	A.T.O.W. 1085 kg	WIND/C H 8 X 0		**VY**	84
Note: Speeds MPH IAS (AFM). Distances AFM fig. 5-4 / 5-5. ASDA = take-off ground roll + landing ground roll. T&G LILE RWY 16: T/O roll 175 m, T/O dist 306 m (TORA 1320, TODA 1320).		**Va**	114	**VFE**	100
		Emax	85	**VLO**	
TORA (Declared) 1420 m	TODA (Declared) 1420 m	ASDA (Declared)		LDA (Declared) 1420 m	
Ground roll 163 m	T/off distance **289 m**	ASDA 326 m		Landing Distance 333 m	

FUEL CALCULATION

DESTINATION Rate	Time in Minutes	Gall./Lt. Required	ALTERNATE Rate	Time in Minutes	Gall./Lt. Required
CLIMB AFM	5	10	CLIMB	-	-
CRUISE 36	46	27	CRUISE 36	32	19
DESC.	-	-	DESC.	-	-
APP. 36	20	13	APP. 36	5	4
Total to Destination		**50**	**Total to Alternate**		**23**
			FINAL RESERVE 30 MIN		19
			TAXI/RUN-UP		3
			MINIMUM FOB		**105**
			ACTUAL FOB		**140**
			EXTRA FOB		**35**

Note:
Fuel in litres (1 L = 0.72 kg).
CLIMB: AFM fig. 5-6, start-up and take-off included.
APP.: 5 min per aerodrome.
CONTINGENCY 15 min, 10 L: included in MINIMUM FOB.
MIN. FOB = DEST + CONT + ALTN + FINAL RES + TAXI.
Whole litres: requirements rounded up.
W&B fuel: take-off fuel 137 L = 99 kg.

LANDING DATA

				FLAP	40°
FIELD ELEV. 1100 ft	RWY 17	RWY COND. ASPHALT, DRY		**Vapp**	70-80
QNH 1017 hPa	O.A.T. 17°C	WIND 180/10 KT		**VFE**	100
M.L.W. 1157 kg	A.L.W. 1049 kg	WIND/C H 10 X 2 R		**VLO**	
Note: Speeds MPH IAS (AFM). Distances AFM fig. 5-4 / 5-5. ASDA = take-off ground roll + landing ground roll. ALTN LIMF RWY 18: LDG roll 167 m, LDG dist 341 m at 1020 kg (LDA 3300). T&G LILE: LDG dist 348 m (LDA 1320).		**Va**	112	**VLE**	
		Emax	85	**Vref Land**	73
TORA (Declared) 1420 m	TODA (Declared) 1420 m	ASDA (Declared)		LDA (Declared) 1420 m	
Ground roll 156 m	T/off distance 257 m	ASDA 297 m		Landing Distance **318 m**	

1

Weather, weight and balance, fuel calculation, take-off and landing data.

4.5 Dashboard – Saved flights

The Dashboard is the first page after signing in. The header (A) greets the pilot by name, with the UTC time, and draws the route network of the home field: the ACV standard routes and the saved flights, with range rings at 25, 50, 100 and 150 NM. The summary tiles (B) give the next or latest flight, the flights ready to fly, those waiting for weather and the planned distance. **New flight** opens the Import page (Section 2). While a flight plan is open in the wizard, **Back to the open flight plan** returns to it. Reloading the page also shows the Dashboard first.

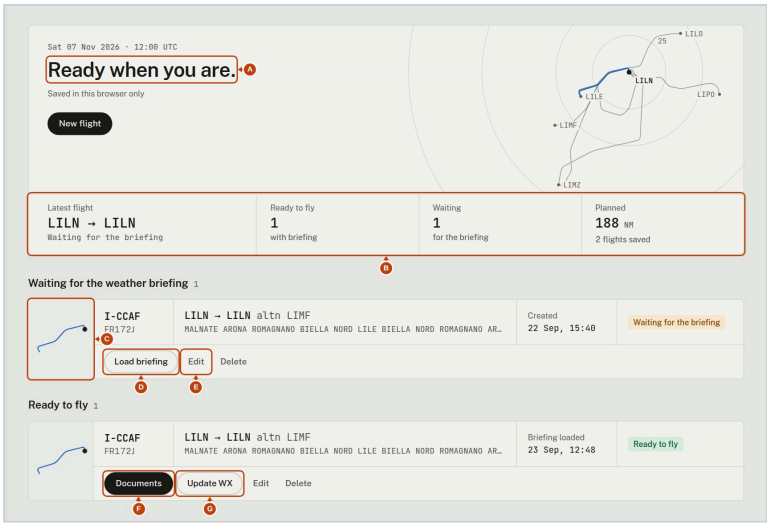
Each saved flight is a strip with the thumbnail of its track (C), drawn from the legs of the Navlog, its badge (STANDBY or READY TO FLY), route, date and times.

1. **SELECT Save** in the wizard to store the flight, for example while waiting for the weather. It appears on the Dashboard as a standby flight.
2. **LOAD** the briefing PDF with **Load briefing** (D) on a standby flight to finalise it: it moves to READY TO FLY.
3. **SELECT Edit** (E) to reopen a flight in the wizard.
4. **SELECT Documents** (F) on a READY TO FLY flight to open its Documents page (Section 4.1).
5. **LOAD** a newer briefing with **Update WX** (G) to update the weather of a READY TO FLY flight.

> **NOTE**
> Opening a saved flight or starting a new one closes the flight plan open in the wizard: with changes not saved, the application asks first whether to save them (Section 3.0).

> **CAUTION**
> Saved flights are stored in your account. On a shared device, select **Sign out** at the end of the session: it removes the local copy from the device. Graphic weather charts are not stored with the flight.

Figure 4.8 – Dashboard



A	**Greeting and route network**	Information
B	**Summary tiles**	Information
C	**Track thumbnail**	VERIFY the route
D	**Load briefing**	Finalise
E	**Edit**	Reopen in the wizard
F	**Documents**	Open the documents
G	**Update WX**	LOAD newer briefing

SECTION 5 Operational Notes – Safety & Performance Limits

5.1 Aircraft applicability

WARNING
AFM data are those of the Reims FR172J (I-CCAF) and FR172H (I-CCAB); empty masses and moments are those of the two aircraft. With **Other**, speeds are those of the FR172J. Used for any other aircraft the results are not approximate: they are wrong.

5.2 Performance limits

ITEM	LIMITATION	EFFECT
Declared distances	TORA / TODA / LDA pre-filled with the physical runway length (OurAirports)	**Must** be corrected from the AIP where stopway, clearway or displaced threshold exist, or NOTAM changes apply
Tailwind	The AFM publishes no tailwind data	Distances are for zero wind and **shorter than actual** (red warning)
Wet / contaminated runway	No AFM data	No increment applied (red warning)
Grass runway	AFM grass increment	Applied when “Grass, dry” is selected
Landing mass	Landing table published at 1157 kg only	Conservative at lower masses
Touch and go	Computed with the take-off mass	Conservative
Headwind credit	Limited to the range of the AFM table / note; mean wind, not the gust	Grey advisory when limited
Gusts	Crosswind limit checked against the gust component	Red warning above 20 kt (T/O) / 15 kt (LDG)
Variable wind (VRB)	No headwind credit; crosswind taken as the full wind or gust speed	Amber advisory: it can also be a tailwind
No runway selected	Wind not applied, distances for zero wind	Amber advisory
V-speeds	AFM tables, linear interpolation by mass (and altitude for VY), no extrapolation	—
Weather for performance	Own METAR; nearest station within 30 NM on request; otherwise ISA standard values	Standard values flagged in red

5.3 Fuel planning as implemented

ITEM	METHOD
Trip	Climb from DEP elevation to cruise altitude (AFM fig. 5-6, 1157 kg, start-up and take-off included) + cruise over the remaining distance at the selected GS and fuel flow. Descent is not modelled separately (included in cruise). + **aerodrome minutes**: 5 min for each leg from or to an aerodrome (first and last leg; the two legs around an en-route aerodrome only if Touch & Go is selected), at cruise fuel flow.
Contingency	max(5% of Trip, 15 min at cruise fuel flow)
Alternate	Cruise, 22 inHg / 2400 RPM at the alternate altitude, over the alternate distance, + 5 aerodrome minutes. No climb allowance.
Final reserve	30 min (VFR) or 45 min (IFR) at cruise fuel flow

ITEM	METHOD
Taxi	(TO − OFF) + (ON − LDG) at 7.5 L/h
Units	1 L = 0.72 kg; 174 L (46 USG) usable

5.4 Navigation data

- Cruise GS is TAS (still air) until corrected for wind by the pilot.
- The alternate distance is a great-circle distance.
- Frequencies are the number only, from OurAirports: CROSS-CHECK with current charts / AIP.
- Navlog times and fuel are recomputed by the application; their totals equal the Trip.

5.5 Daylight check

- Sunrise / sunset: upper limb at −0.833°; civil twilight: centre at −6° (night as defined in Regulation (EU) 923/2012).
- Rounded conservatively: sunset and end of twilight to the minute before, sunrise to the minute after.
- Uses the flight date of Step 8 and the UTC times of Step 1. No elevation correction (as in published tables).

5.6 Documents and storage

- The documents are downloaded from the Documents page and printed from the PDF, with the PDF viewer.
- They are generated from the flight data at the moment of the download: after any change, download them again.
- Saved flights are stored in the pilot's account and are visible only to that pilot. On a shared device, **Sign out** at the end of the session.
- The A5 NAV-FLIGHTPLAN and the club navlog are separate documents, never part of the OFP.

The documents produced are an aid to planning. Responsibility for the flight remains with the Pilot in Command.

APPENDIX A Pre-Departure Cross-Check

Carry out before printing the documents. Each item refers to the section where the check is described.

ITEM	CHECK	REF.	✓
Times	OFF / TO in UTC; LDG, ON, BLOCK, FLT plausible	3.1	☐
Daylight	TO, LDG and ALTN landing reviewed; no unaccepted Civil twilight / Night	3.1	☐
Cruise	Altitude, RPM, MAP selected; GS corrected for wind	3.2	☐
Climb	Cruise altitude reached; TOC position plausible	3.2	☐
Alternate	ALTN entered; distance and altitude verified	3.2	☐
Fuel	Plan Block within 174 L usable; FOB not INSUFFICIENT	3.2	☐
Mass & balance	Masses within limits; CG within envelope at take-off and both landings	3.3	☐
Runways	Runway selected on every tab; TORA / TODA / LDA cross-checked with AIP and NOTAM	3.4	☐
Performance	All margins positive; every Checks & Warnings item reviewed	3.4	☐
Navlog	Totals equal Trip time and fuel; frequencies verified	3.5	☐
Weather & NOTAM	Correct aerodrome and current issue in every box; SIGMET / AIRMET box not empty	3.8	☐
Date / PIC	Flight date and PIC name entered	3.8	☐
Profile	Vertical profile without amber legs, or each one accepted	3.5	☐
Documents	OFP and navlogs downloaded from the Documents page after the last change; OFP printed and signed	4.1	☐